

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: DispY-Rev-CI.sil etc.

Creation date: 4-Nov-83 11:39:26

For: Spruce

33 total sheets = 32 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

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Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

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Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

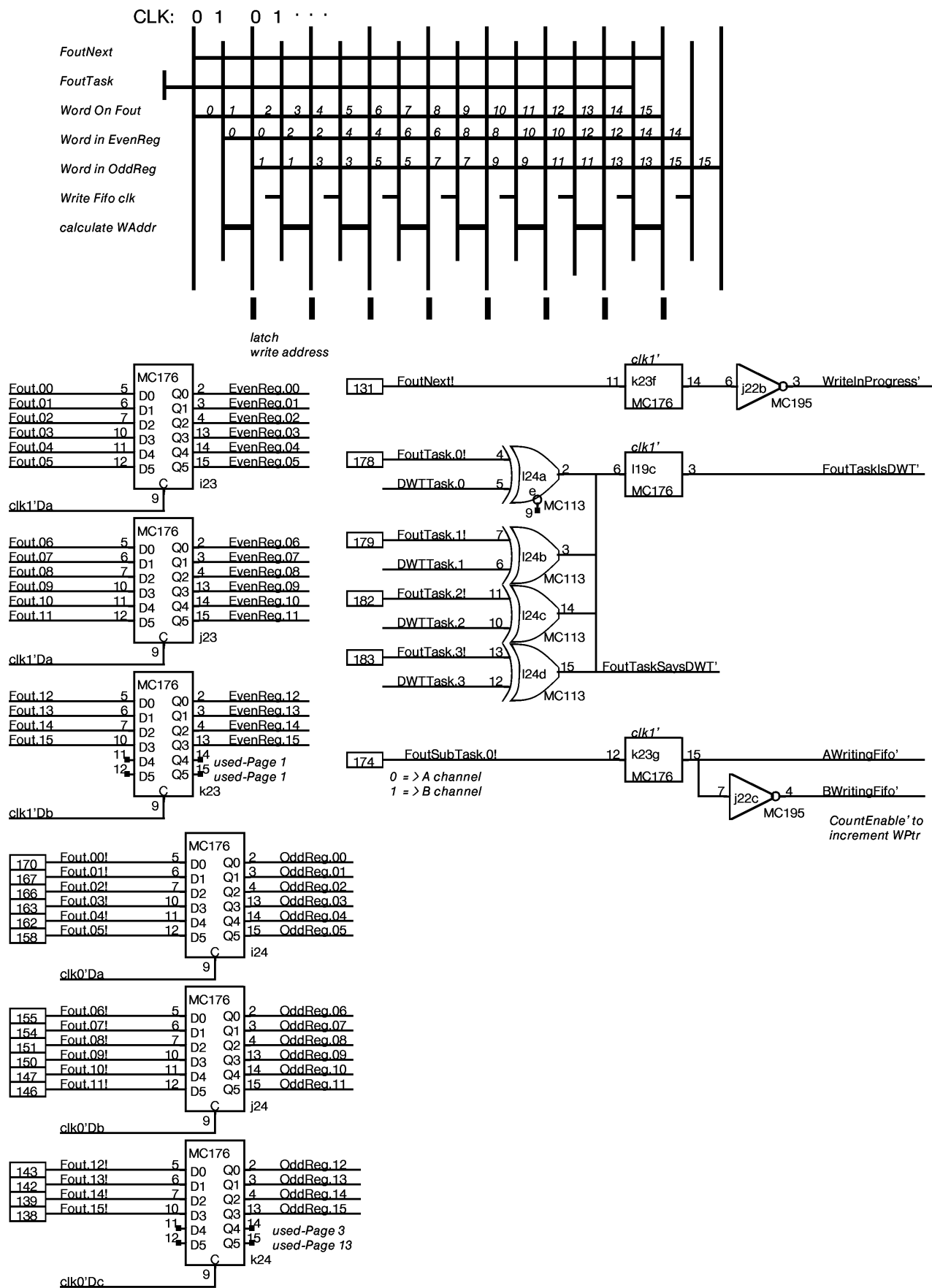
Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

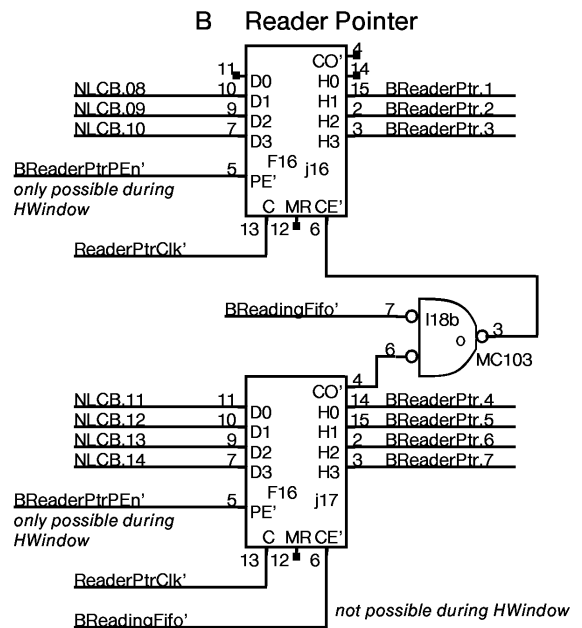
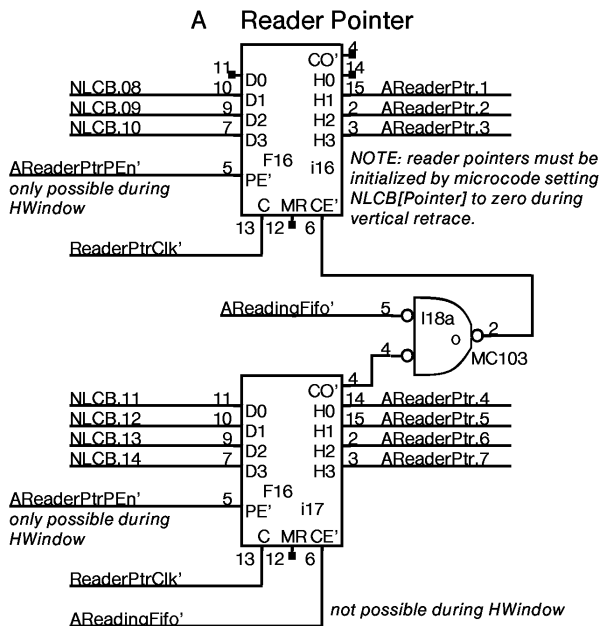
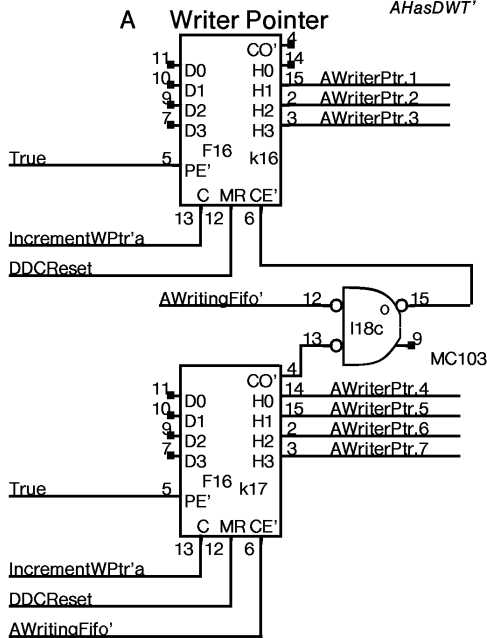
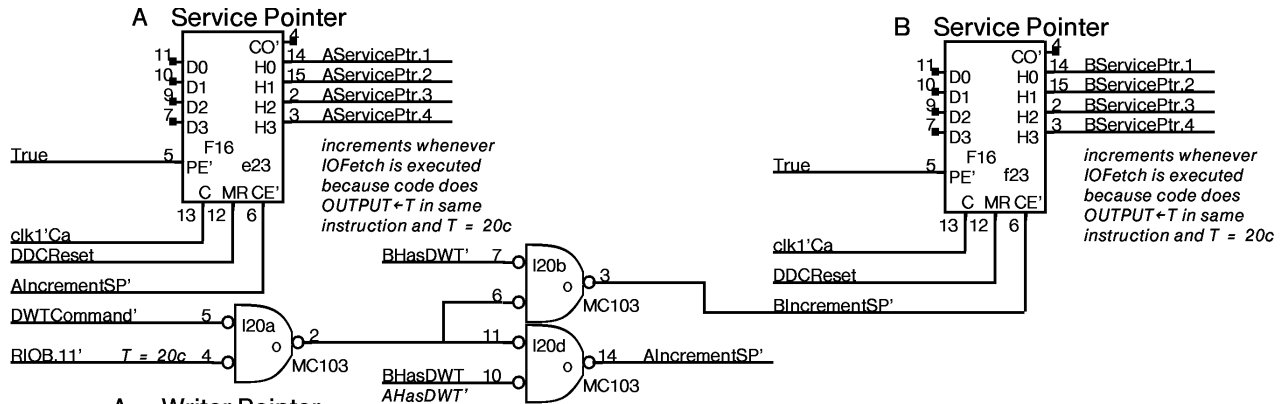
DORADO SCHEMATICS

Display Y

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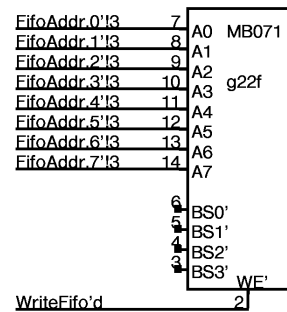
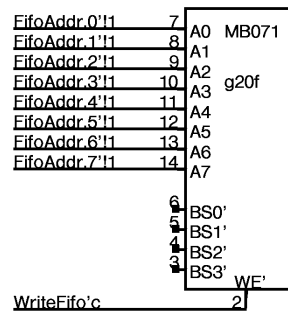
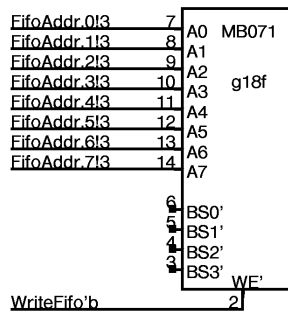
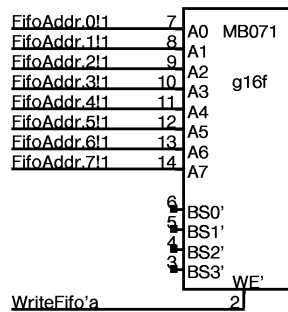


EvenReg.00	19	g16b	23	Fifo.00
		MB071		
EvenReg.01	18	g16c	22	Fifo.01
		MB071		
EvenReg.02	17	g16d	21	Fifo.02
		MB071		
EvenReg.03	16	g16e	20	Fifo.03
		MB071		
EvenReg.04	19	g18b	23	Fifo.04
		MB071		
EvenReg.05	18	g18c	22	Fifo.05
		MB071		
EvenReg.06	17	g18d	21	Fifo.06
		MB071		
EvenReg.07	16	g18e	20	Fifo.07
		MB071		
EvenReg.08	19	g20b	23	Fifo.08
		MB071		
EvenReg.09	18	g20c	22	Fifo.09
		MB071		
EvenReg.10	17	g20d	21	Fifo.10
		MB071		
EvenReg.11	16	g20e	20	Fifo.11
		MB071		
EvenReg.12	19	g22b	23	Fifo.12
		MB071		
EvenReg.13	18	g22c	22	Fifo.13
		MB071		
EvenReg.14	17	g22d	21	Fifo.14
		MB071		
EvenReg.15	16	g22e	20	Fifo.15
		MB071		

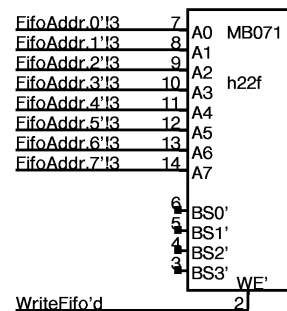
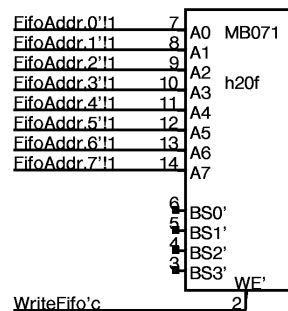
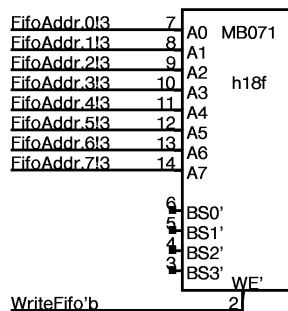
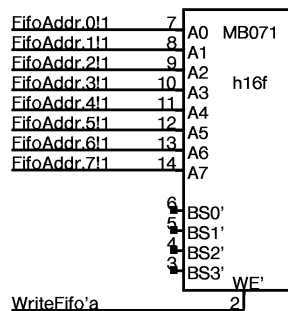
EVEN WORDS

OddReg.00	19	h16b	23	Fifo.16
		MB071		
OddReg.01	18	h16c	22	Fifo.17
		MB071		
OddReg.02	17	h16d	21	Fifo.18
		MB071		
OddReg.03	16	h16e	20	Fifo.19
		MB071		
OddReg.04	19	h18b	23	Fifo.20
		MB071		
OddReg.05	18	h18c	22	Fifo.21
		MB071		
OddReg.06	17	h18d	21	Fifo.22
		MB071		
OddReg.07	16	h18e	20	Fifo.23
		MB071		
OddReg.08	19	h20b	23	Fifo.24
		MB071		
OddReg.09	18	h20c	22	Fifo.25
		MB071		
OddReg.10	17	h20d	21	Fifo.26
		MB071		
OddReg.11	16	h20e	20	Fifo.27
		MB071		
OddReg.12	19	h22b	23	Fifo.28
		MB071		
OddReg.13	18	h22c	22	Fifo.29
		MB071		
OddReg.14	17	h22d	21	Fifo.30
		MB071		
OddReg.15	16	h22e	20	Fifo.31
		MB071		

ODD WORDS

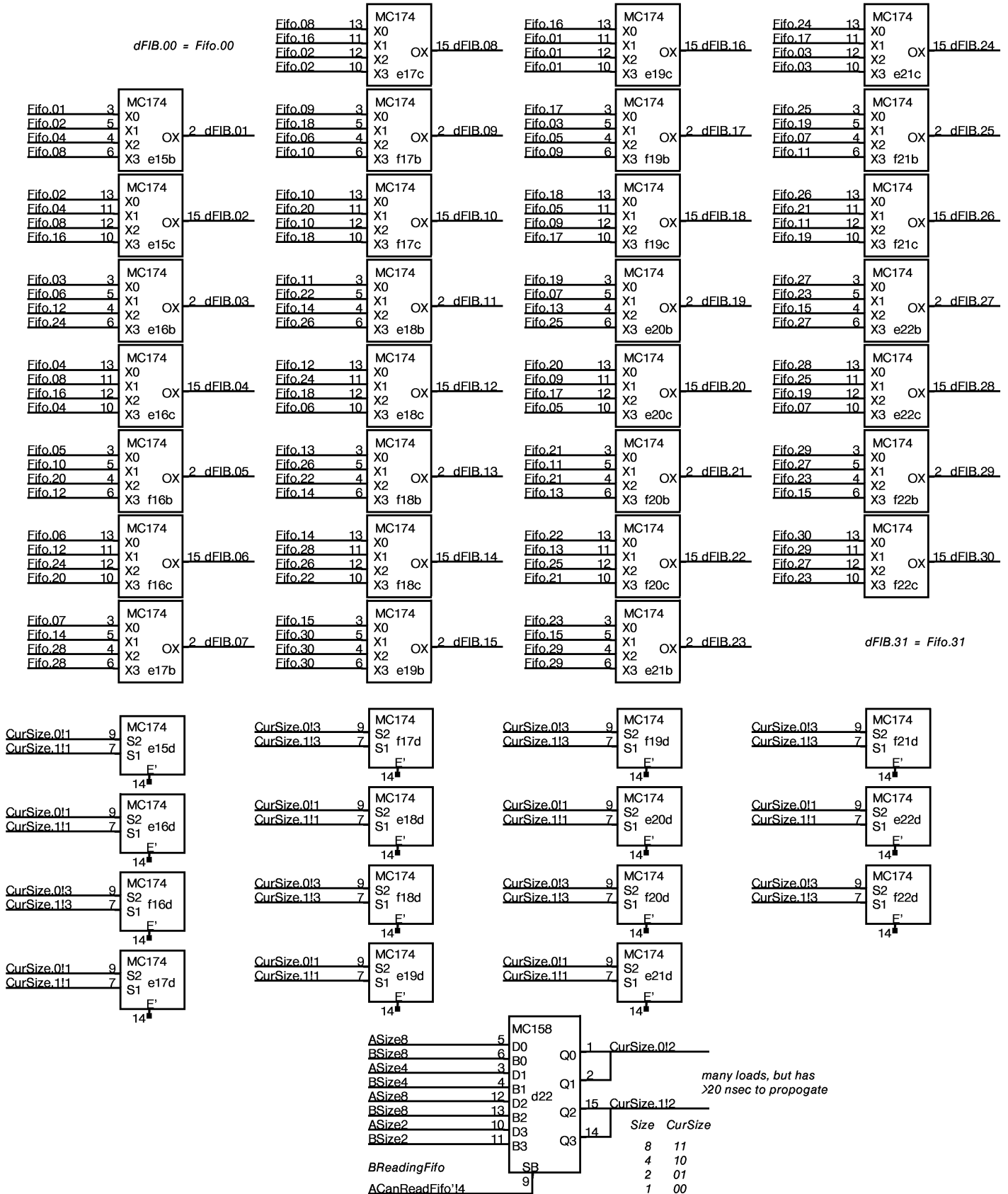


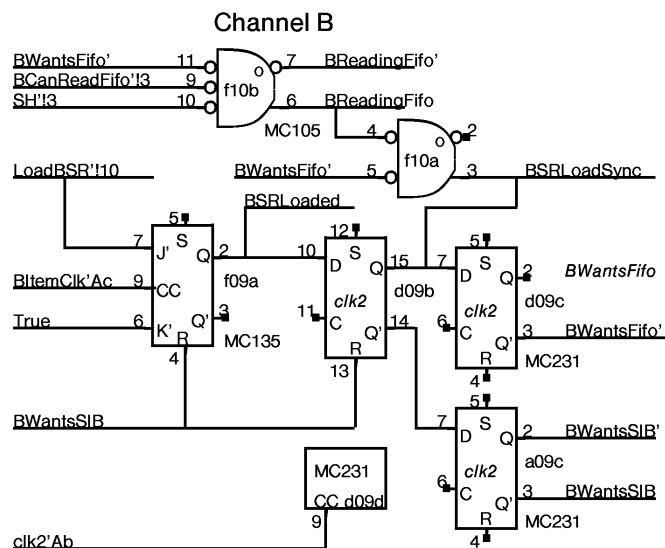
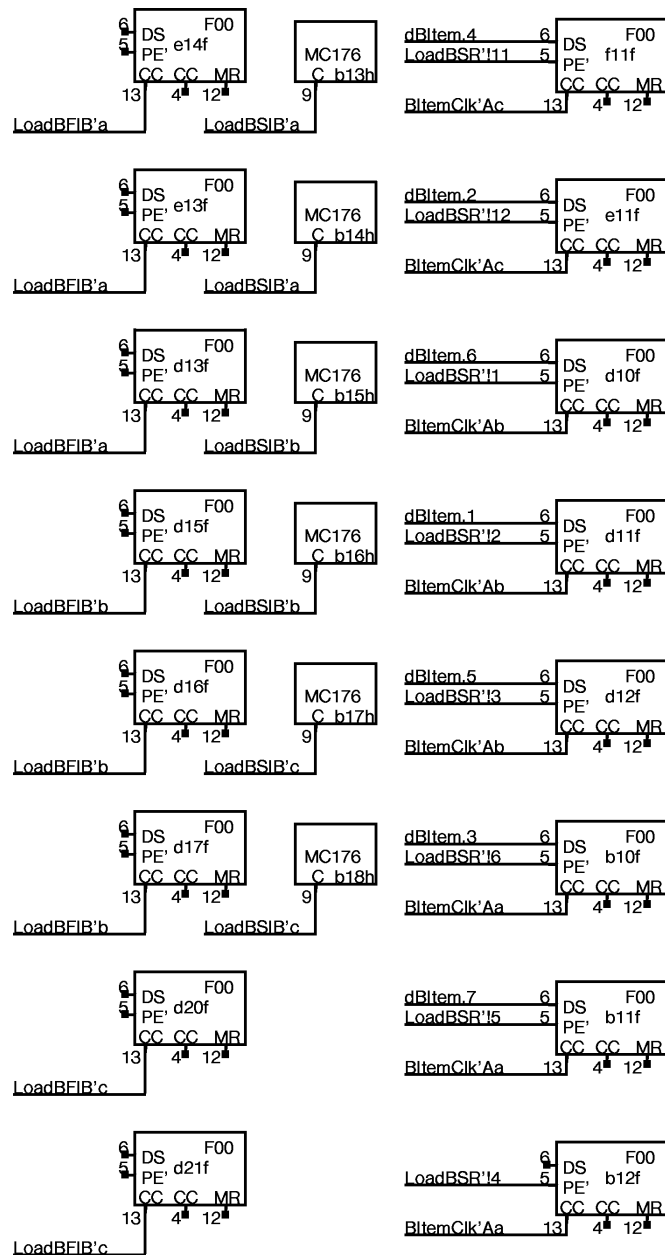
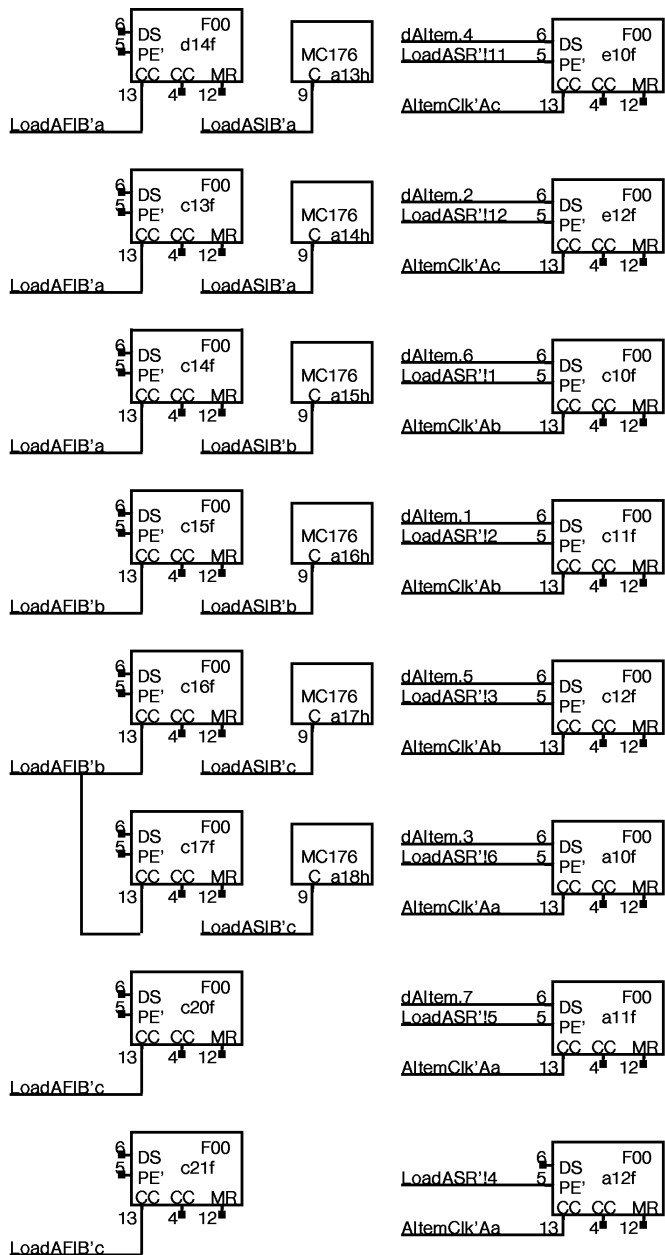
EVEN WORDS



ODD WORDS

See ItemGenerator table for input specs



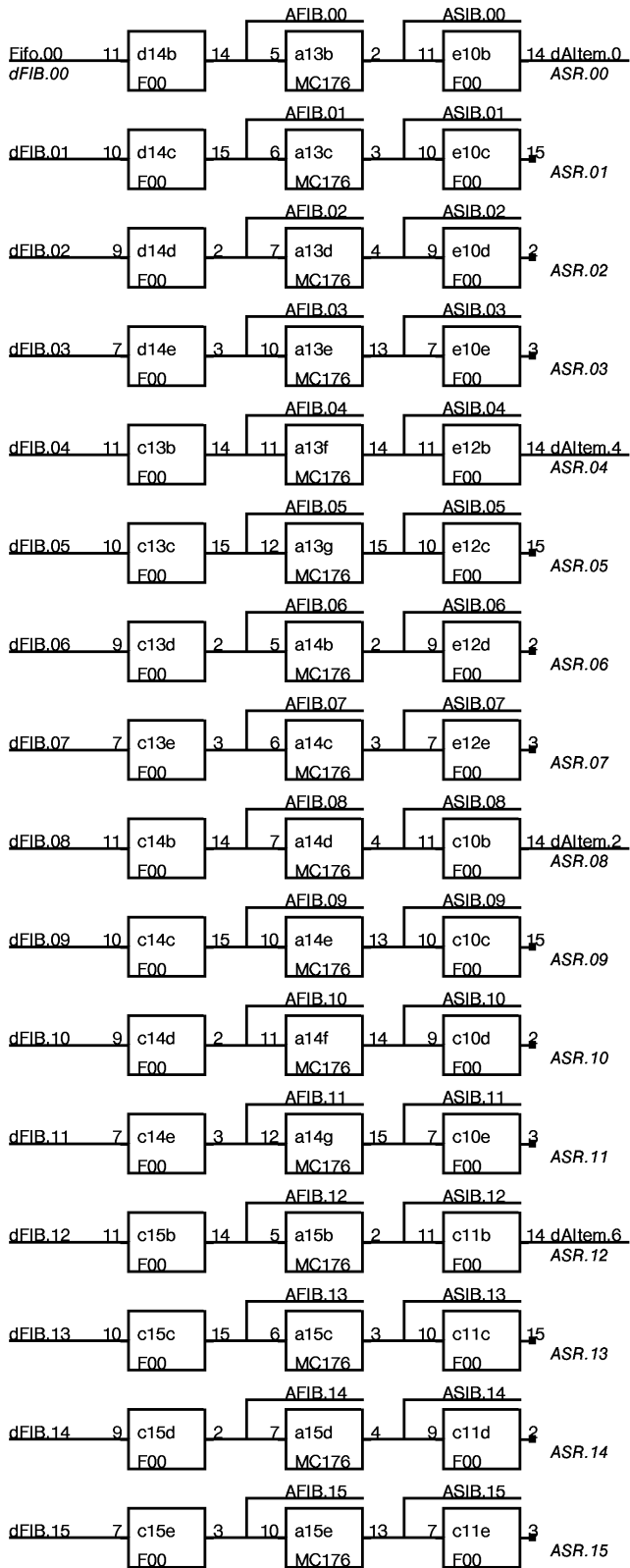


Channel A

Reworked synchronizer

moved to page 8, Rev Ck

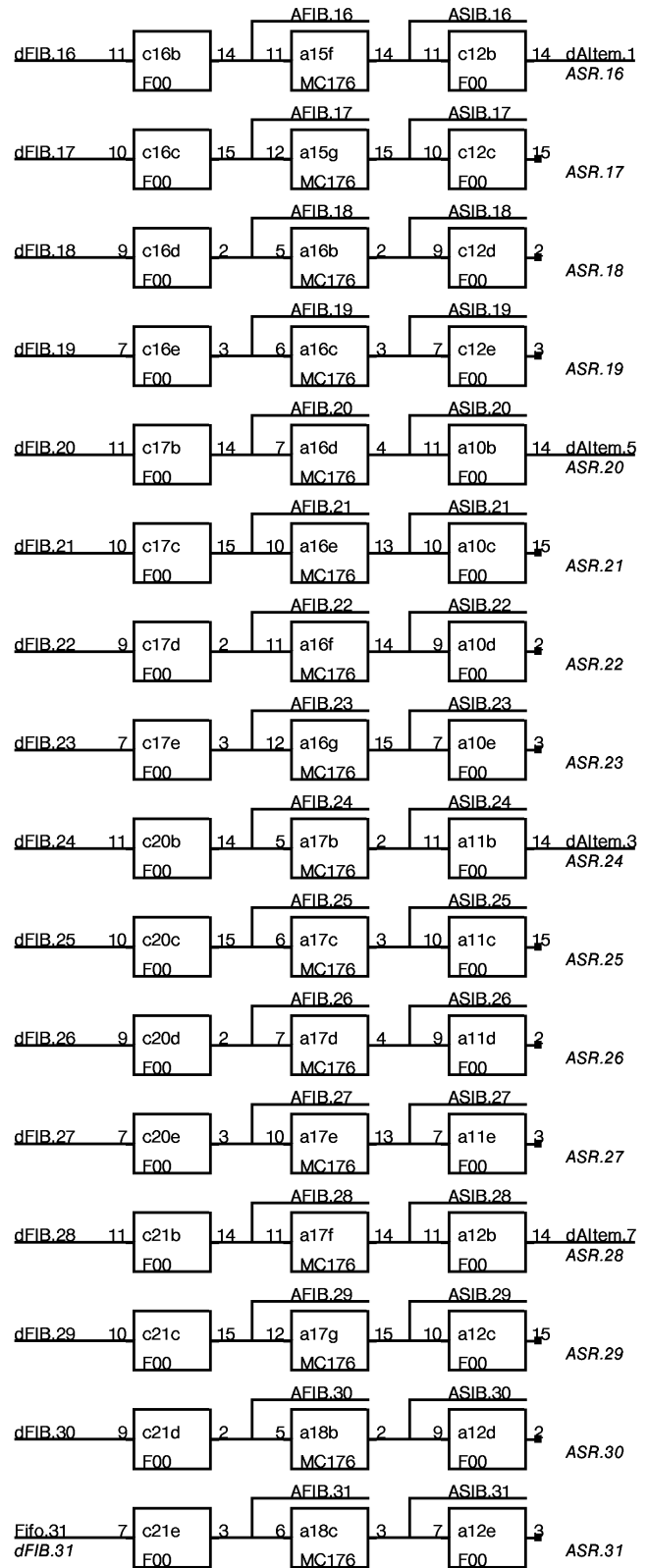
Channel B



AFIB

ASIB

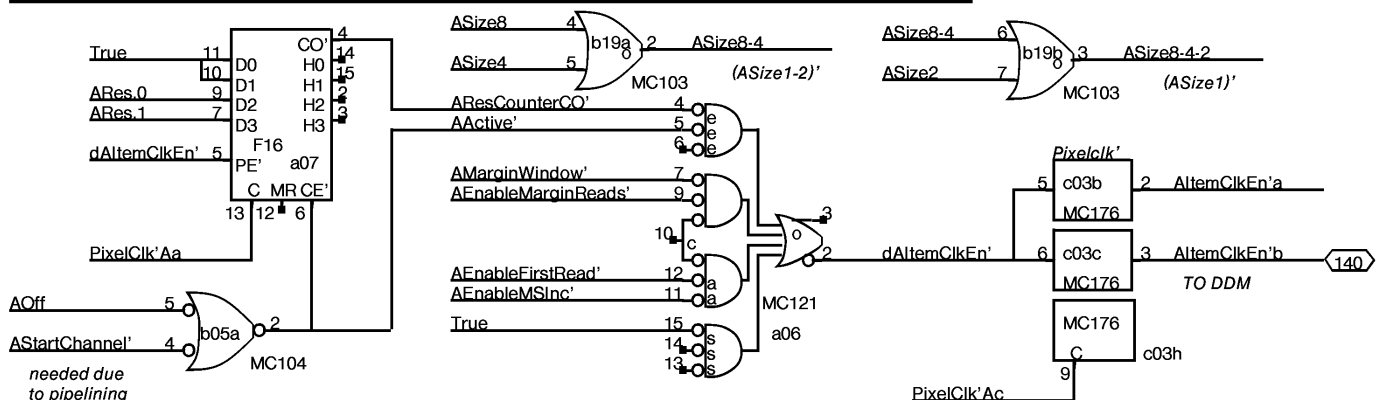
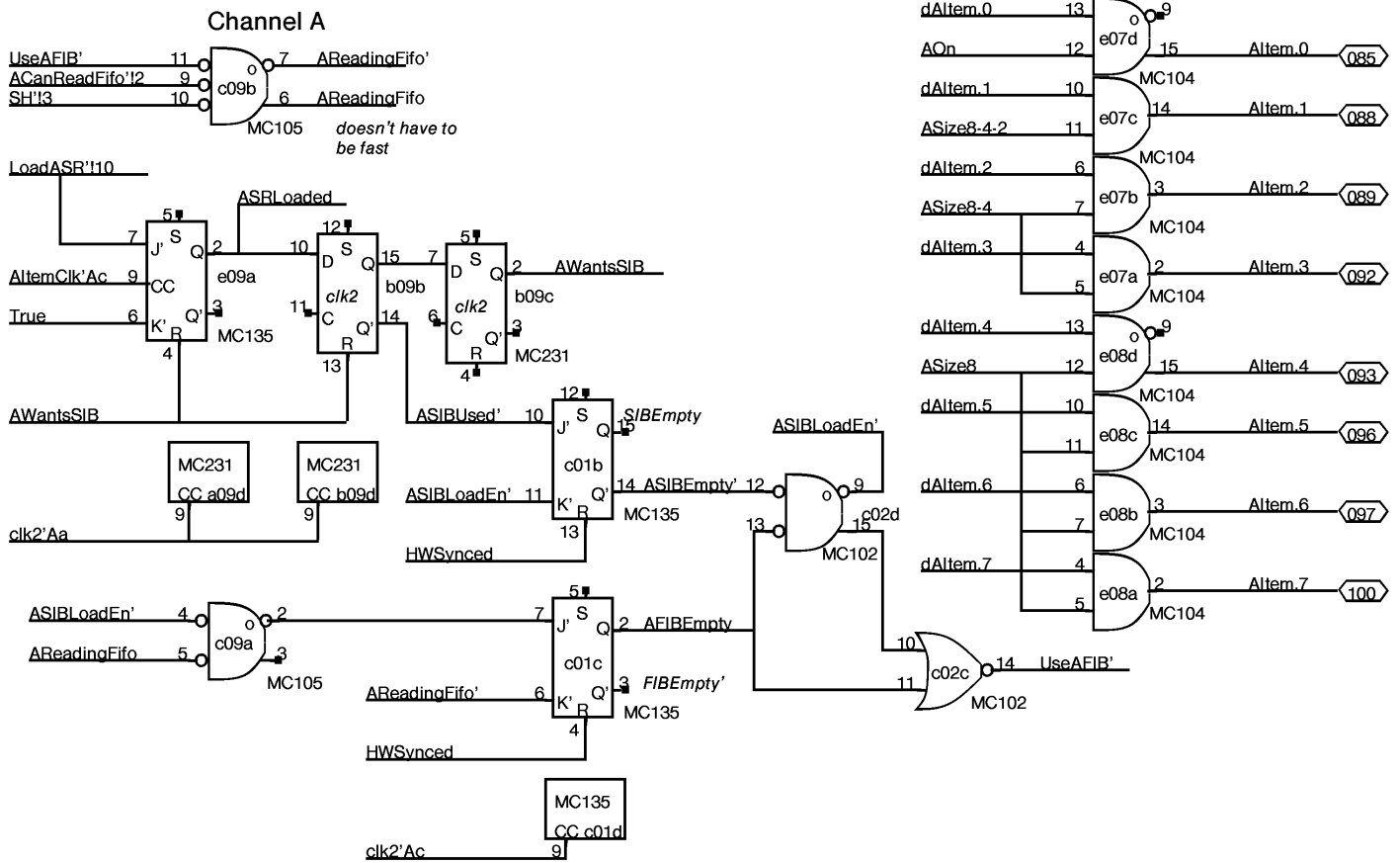
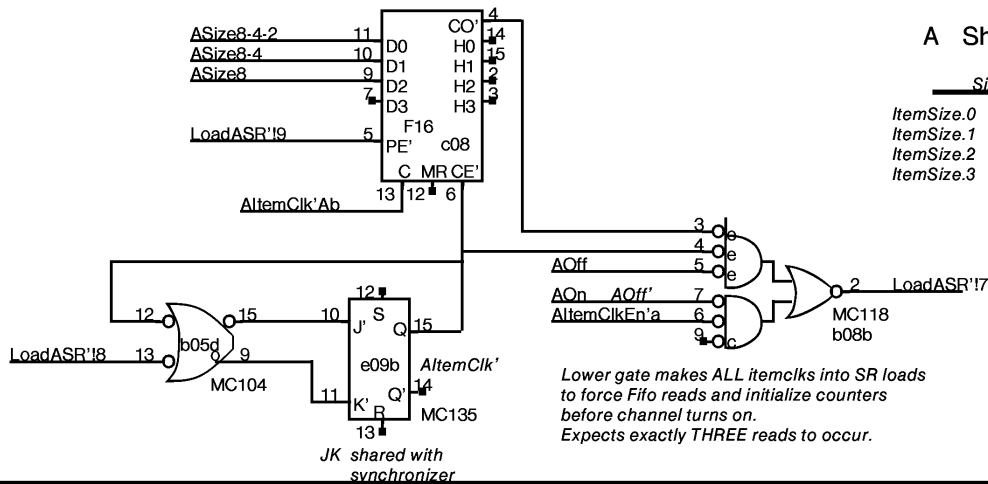
ASR

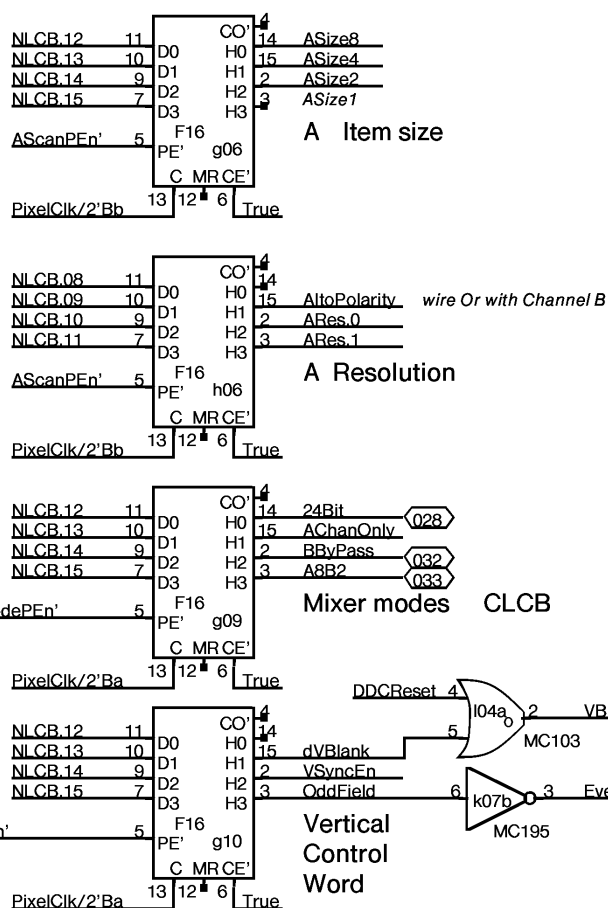
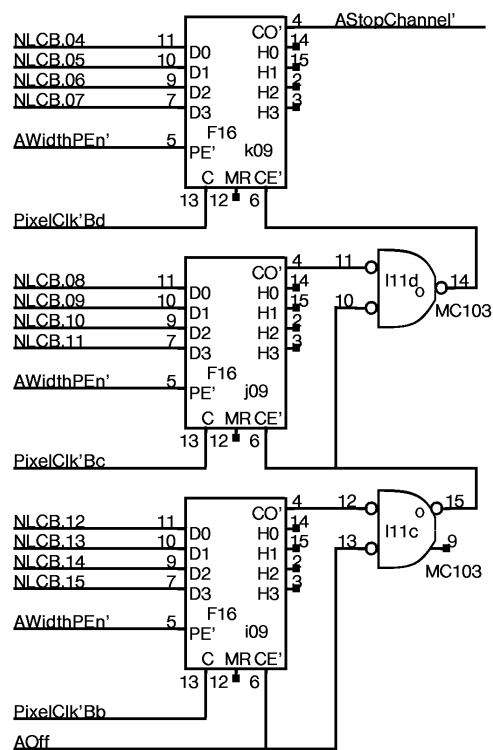
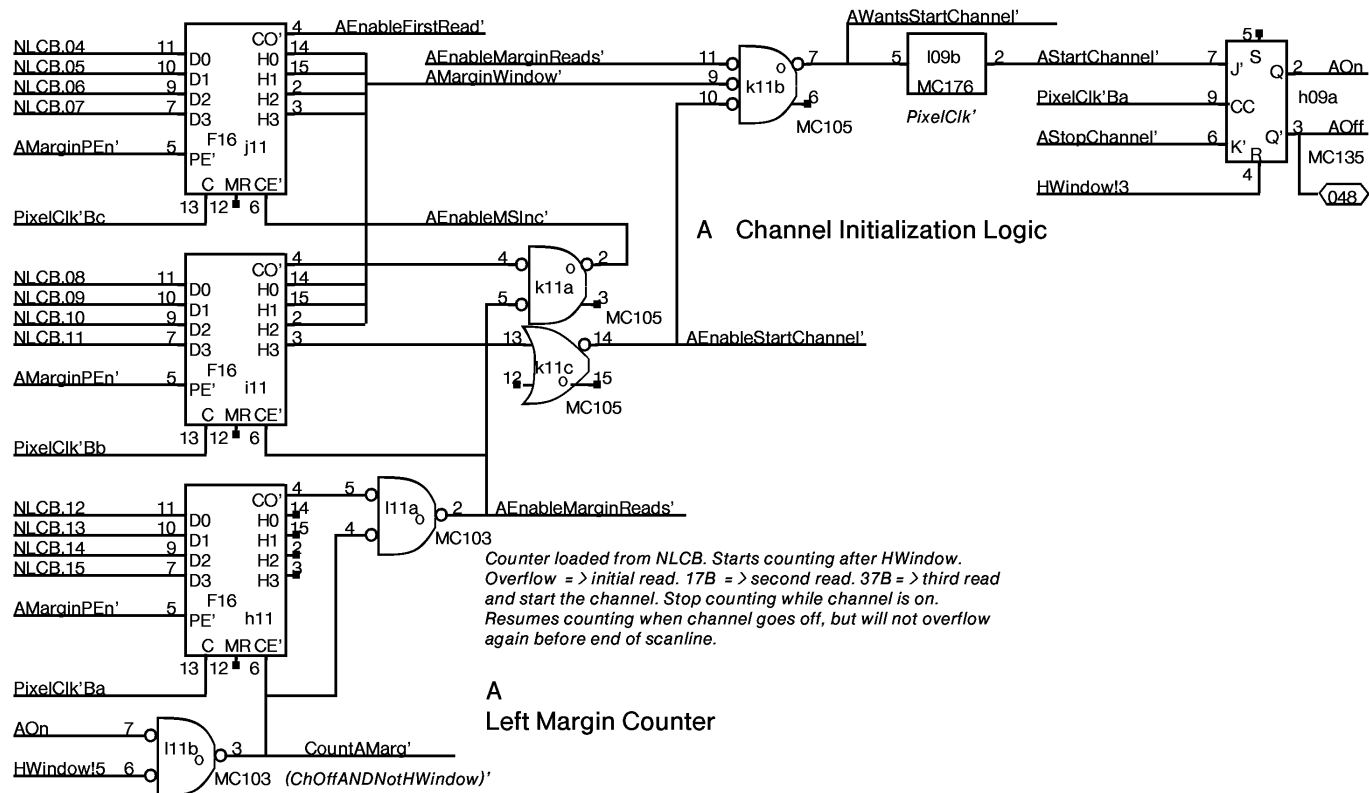


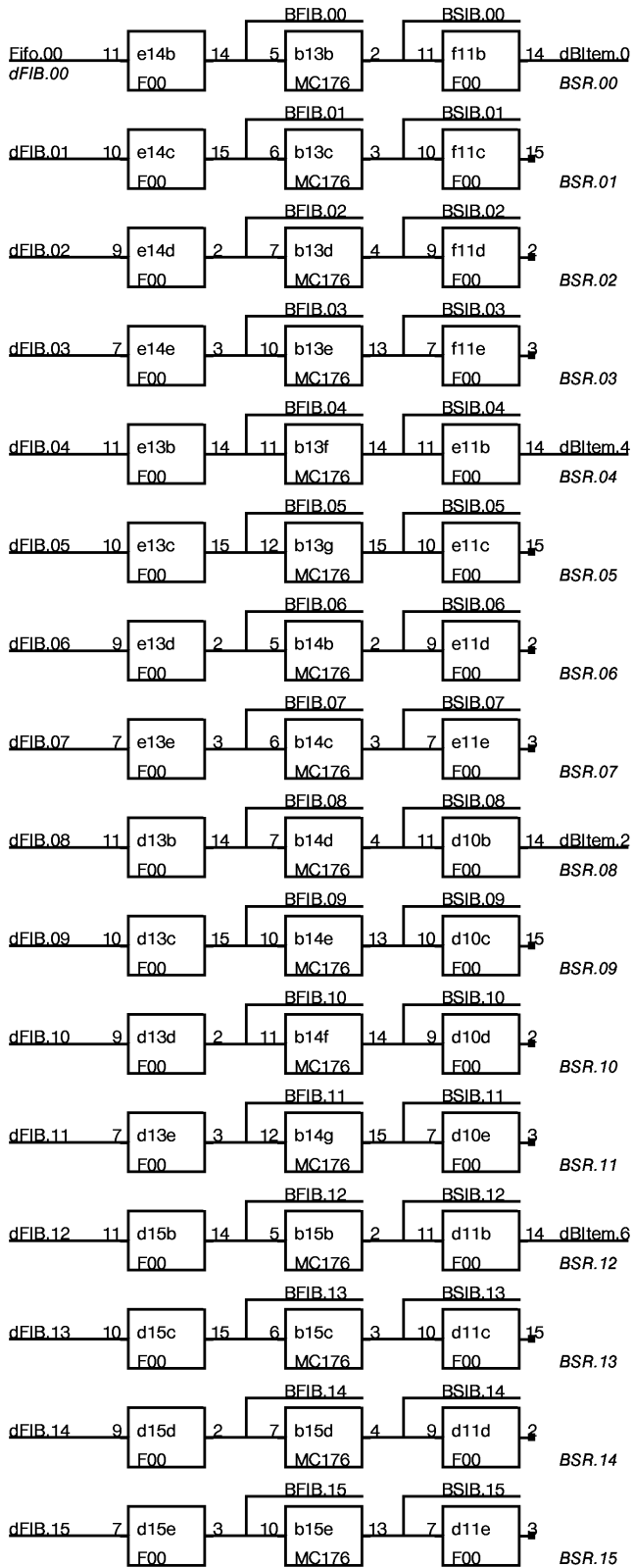
AFIB

ASIB

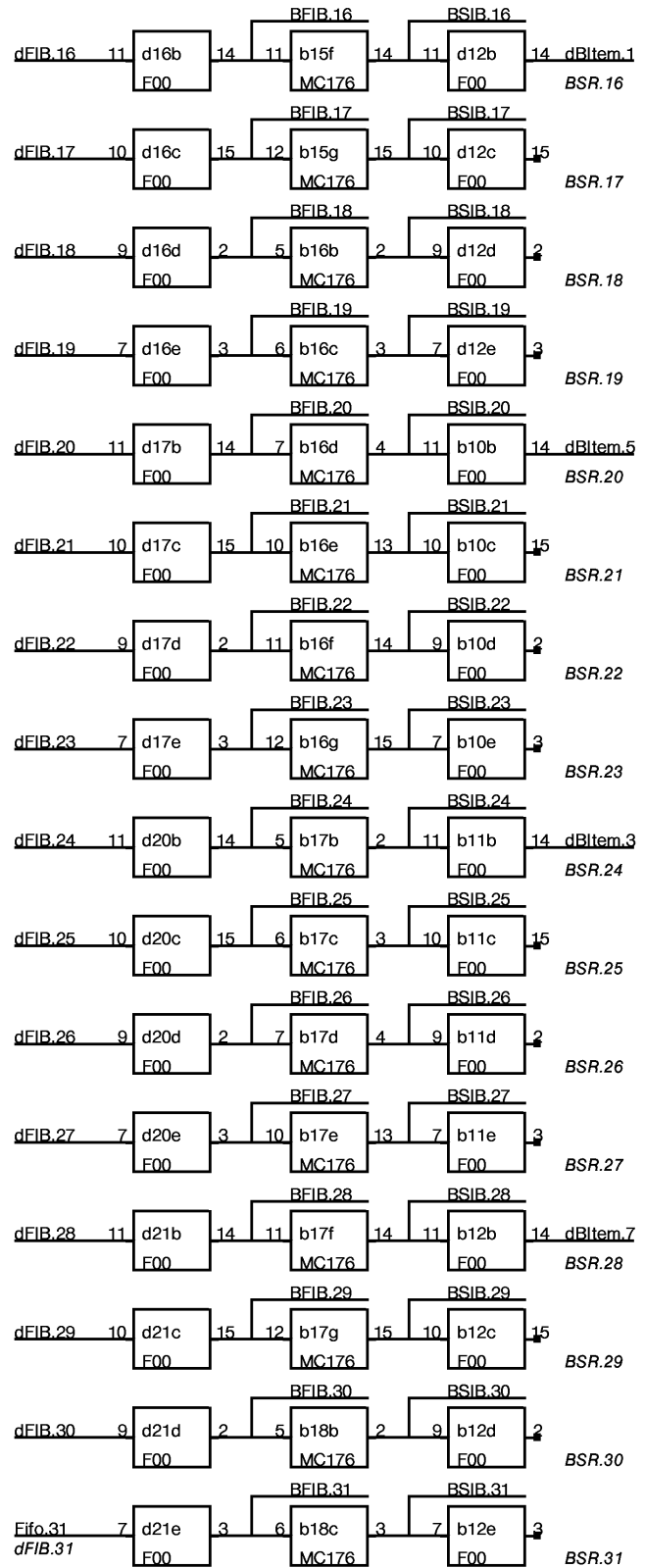
ASR



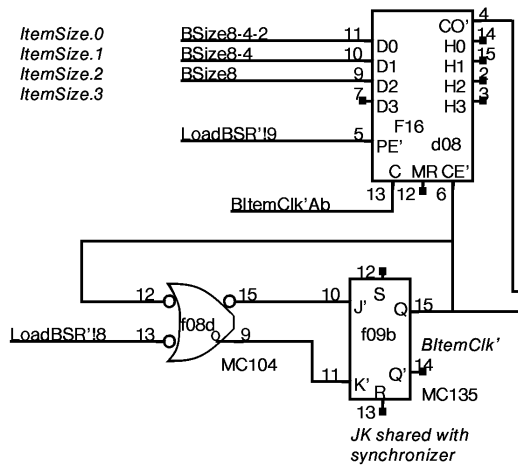




BFIB BSIB BSR



BFIB BSIB BSR

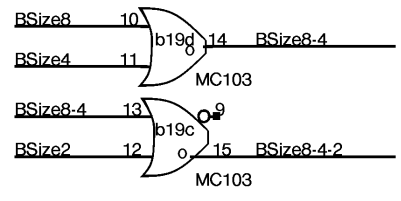


B Shift Register Control

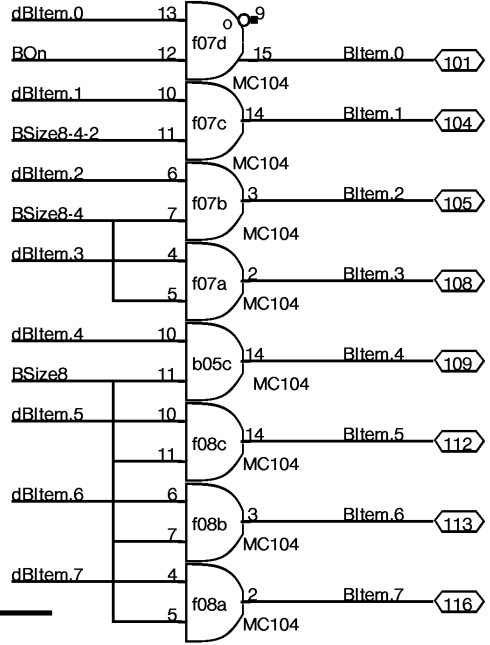
Size	8	4	2	1
ItemSize.0	1	1	1	0
ItemSize.1	1	1	0	0
ItemSize.2	1	0	0	0
ItemSize.3	0	0	0	0

Upper term makes ALL ItemClks into BSR loads to force Fifo reads and initialize counters before channel turns on. Expects exactly THREE reads to occur.

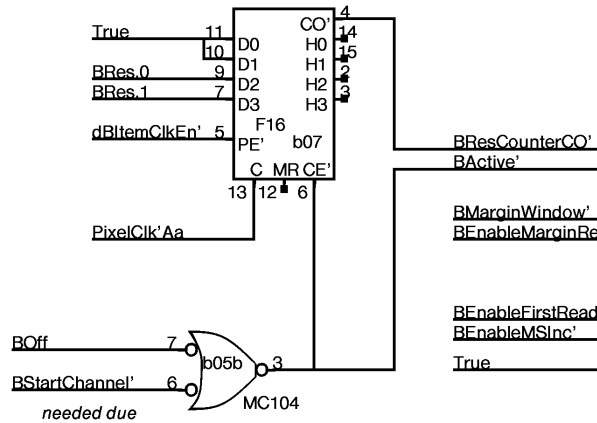
B Item buffer logic



dItem are garbage until channel becomes active



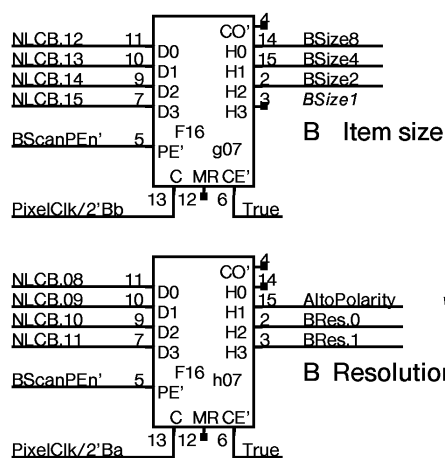
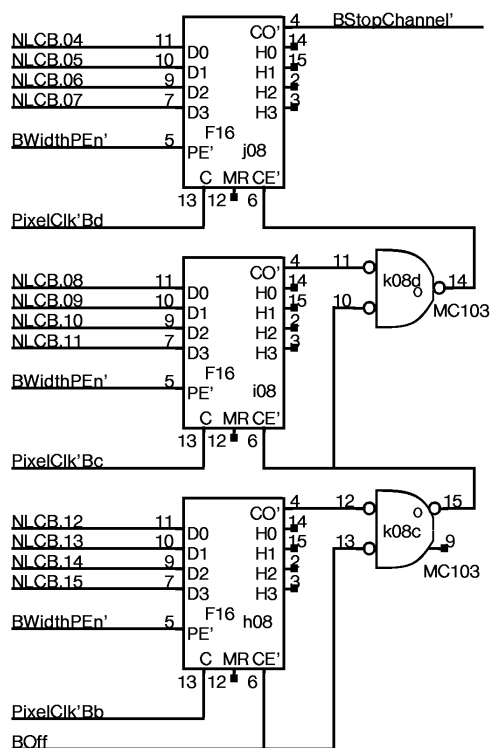
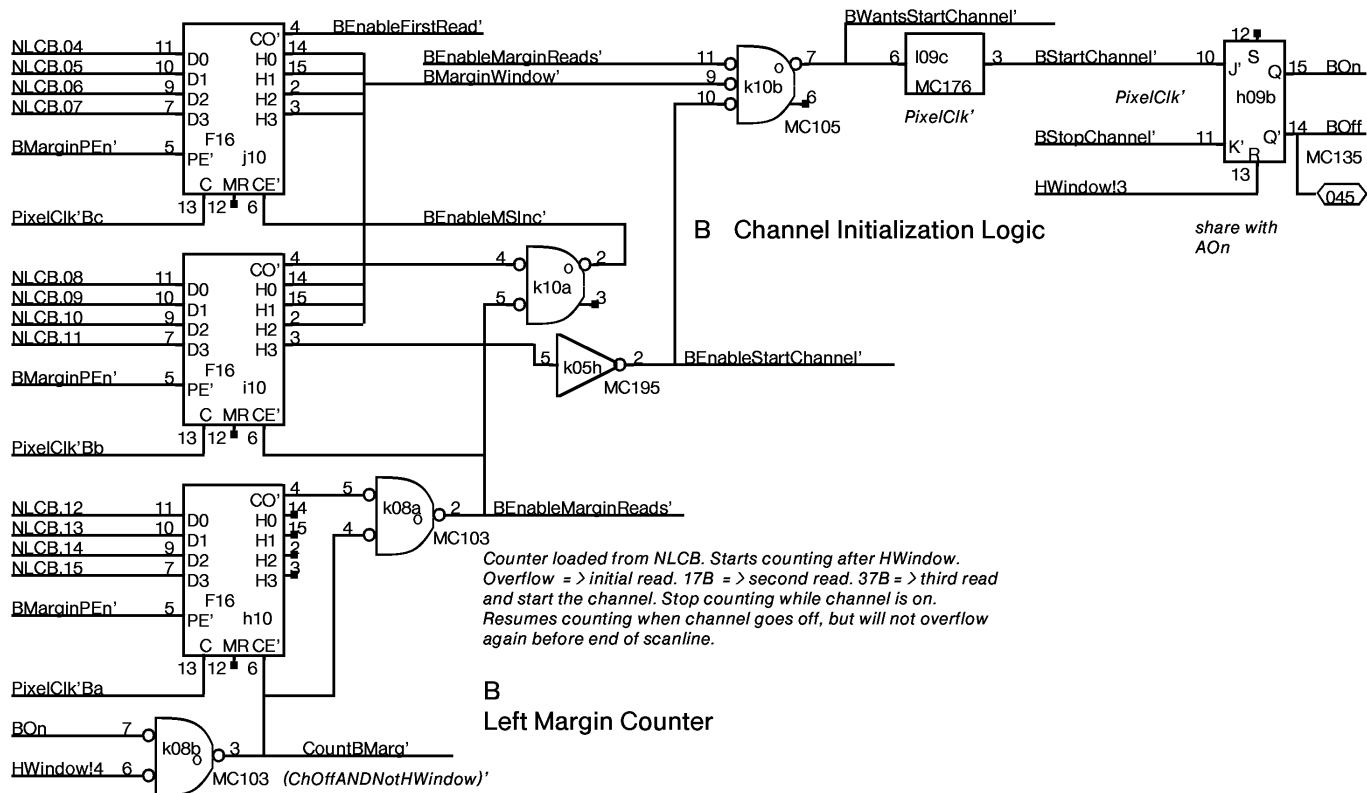
Items are ignored until channel turns ON by master reset of HOLD registers on DDM

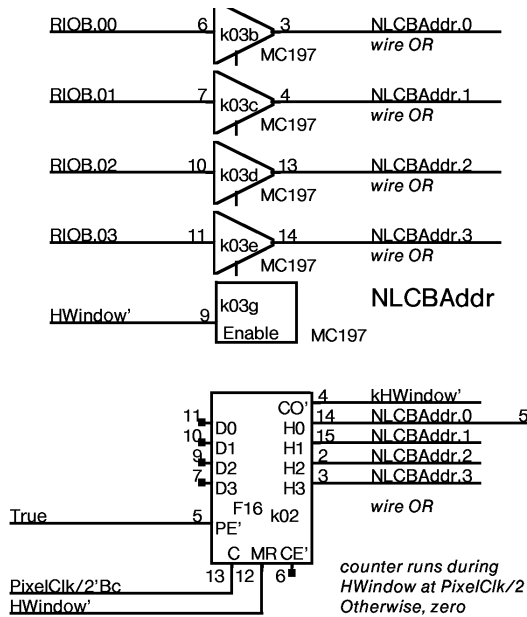


B Item Clock Generator

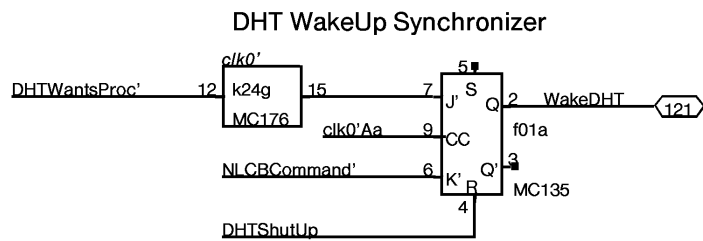
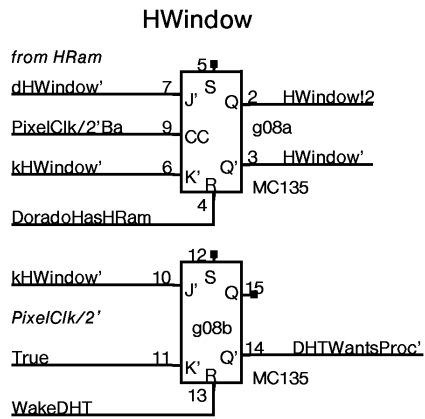
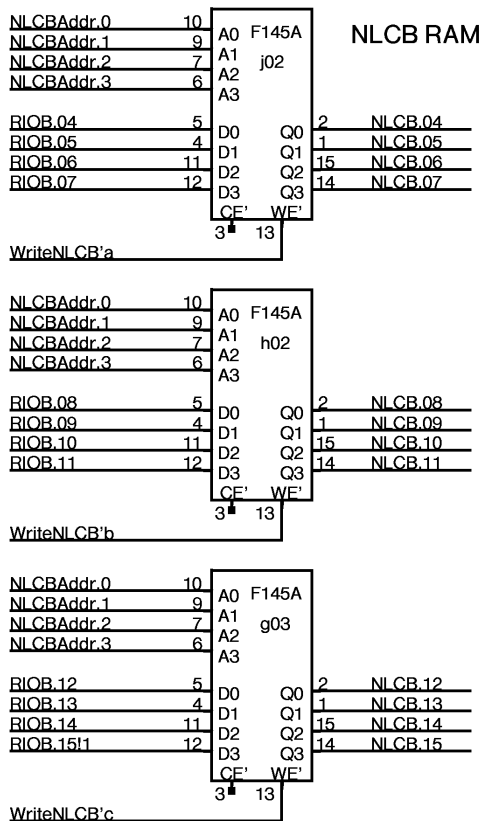
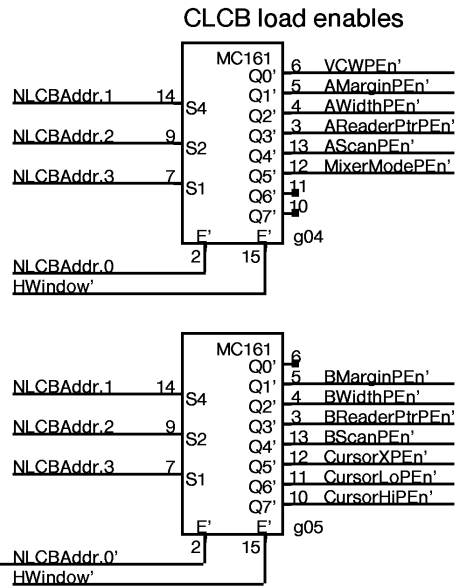
Resolution	Full	Half	Quarter
True	1	1	1
True	1	1	1
Res.0	1	1	0
Res.1	1	0	0

Force exactly 3 ItemClks to occur in order to initialize channel data and clk counters. The third forced ItemCk loads SR with good data and turns the channel ON.





NOTE: PC/2 only valid
 while HRam address
 counter is incrementing



For convenience, use NLCB command
 to kill wakeup. Assumes DHT will always
 do some NLCB command whenever it is
 awakened. Default would be to load an
 unused, don't care address in NLCB.

requires minimum 4 instruction loop

THE DDC MAINTAINS, FOR EACH CHANNEL, TWO FLAGS:

NEXT Word Control Block Flag (NextWCBFlag)
Current Word Control Block Flag (CurrentWCBFlag)

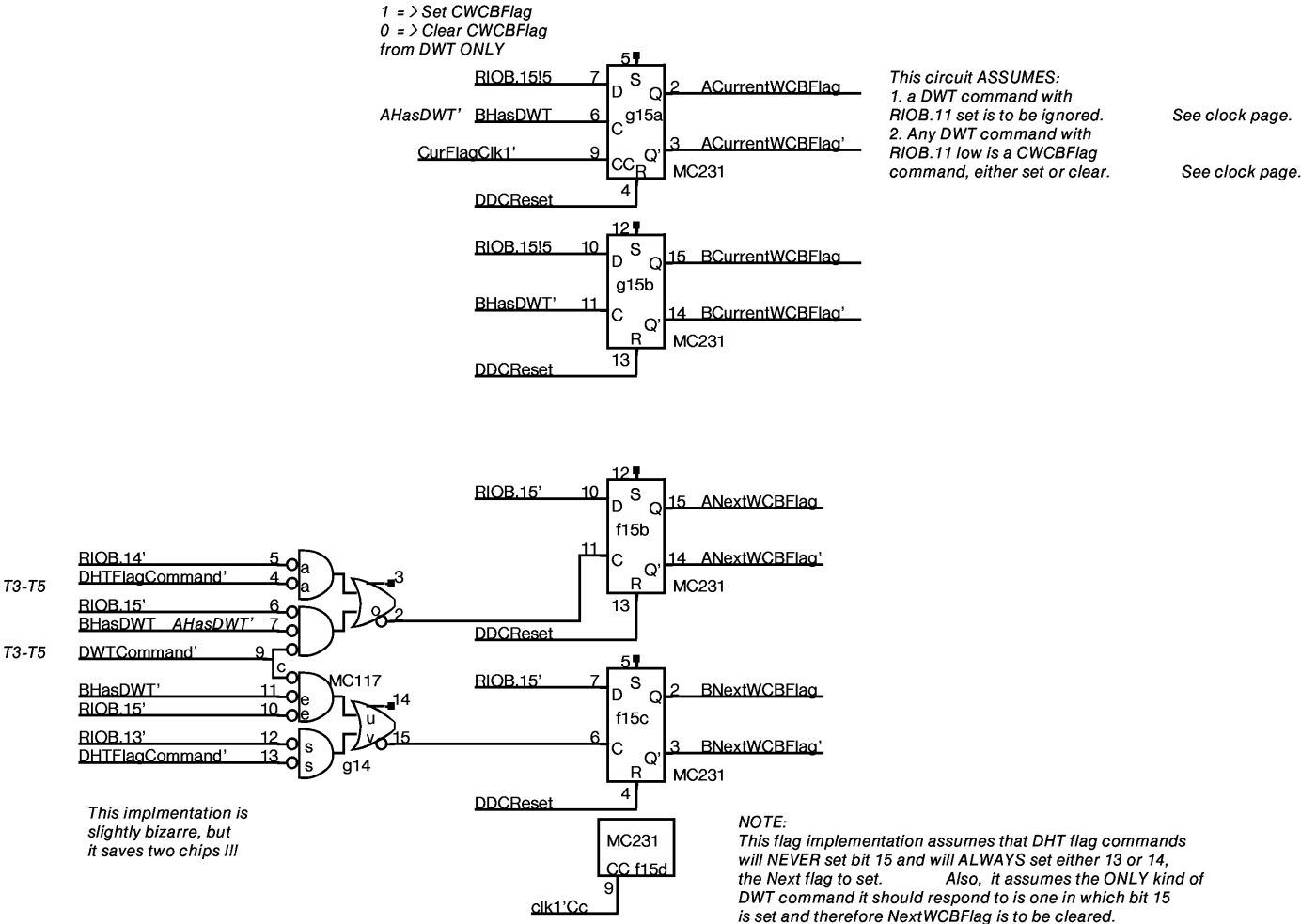
A Word Control Block is a pair of values called Address and MunchCount, for either the CURRENT or the NEXT scanline.

Flag management	SET	CLEARED
NextWCBFlag	by DHT when it has filled the NextWCB	by DWT when it has copied NextWCB into CurrentWCB
CurrentWCBFlag	by DWT when it has copied NextWCB into CurrentWCB	by DWT after it has sent out all the data from the CurrentWCB

WakeUp conditions:

WakeDHT: whenever CLCB ← NLCB i.e.: end of every HWindow

WakeDWT: (CurrentWCBFlag AND BufferAvailable) OR (NextWCBFlag AND NOT(CurrentWCBFlag))

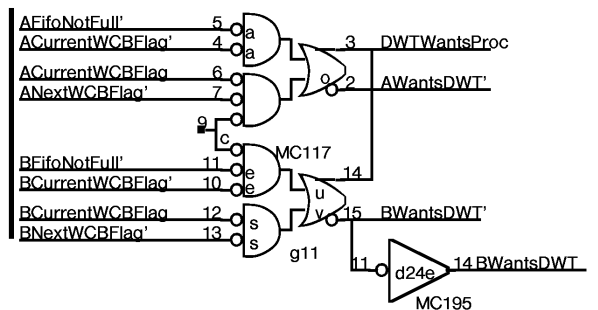
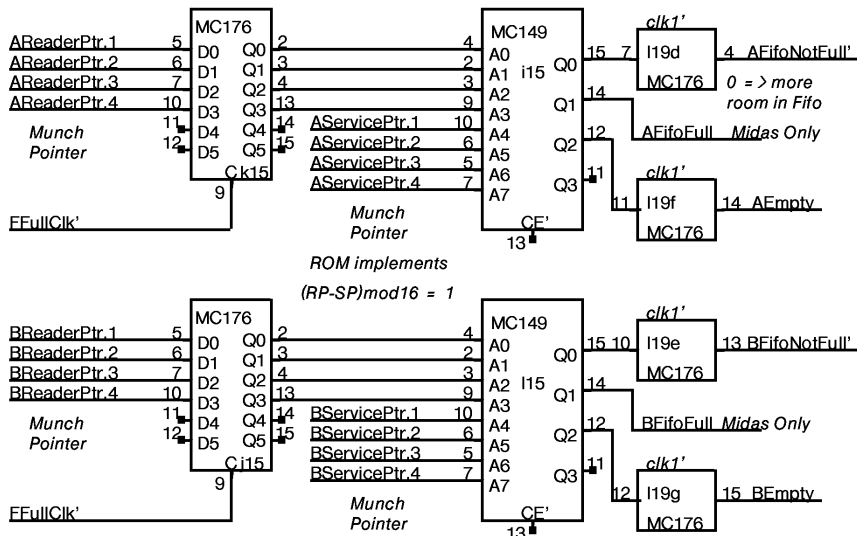
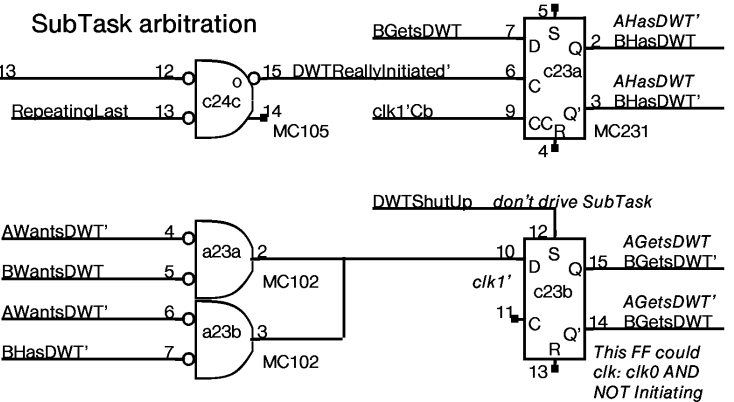
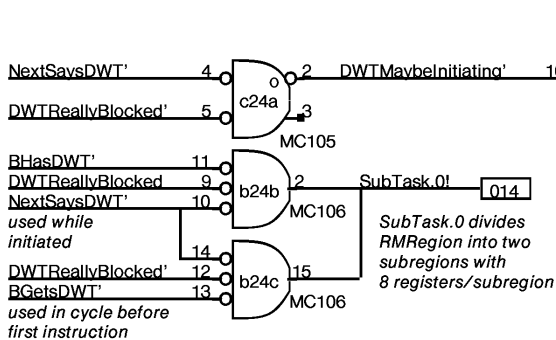
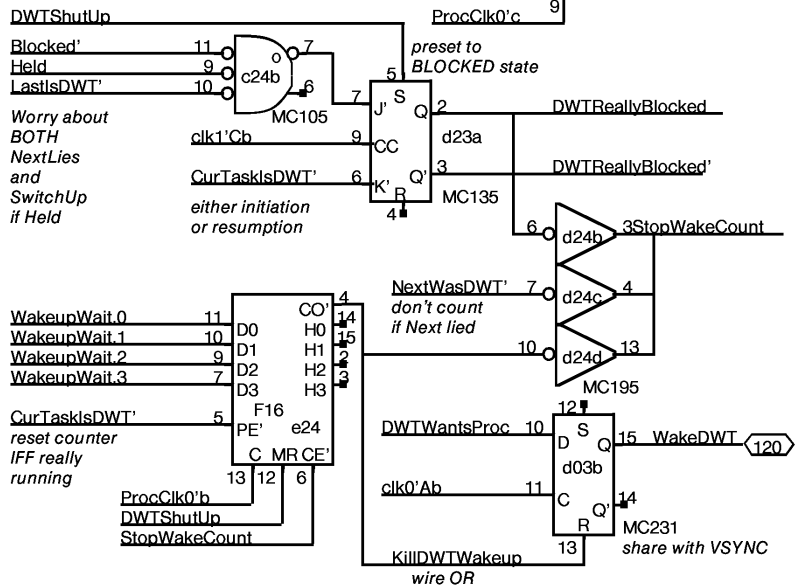
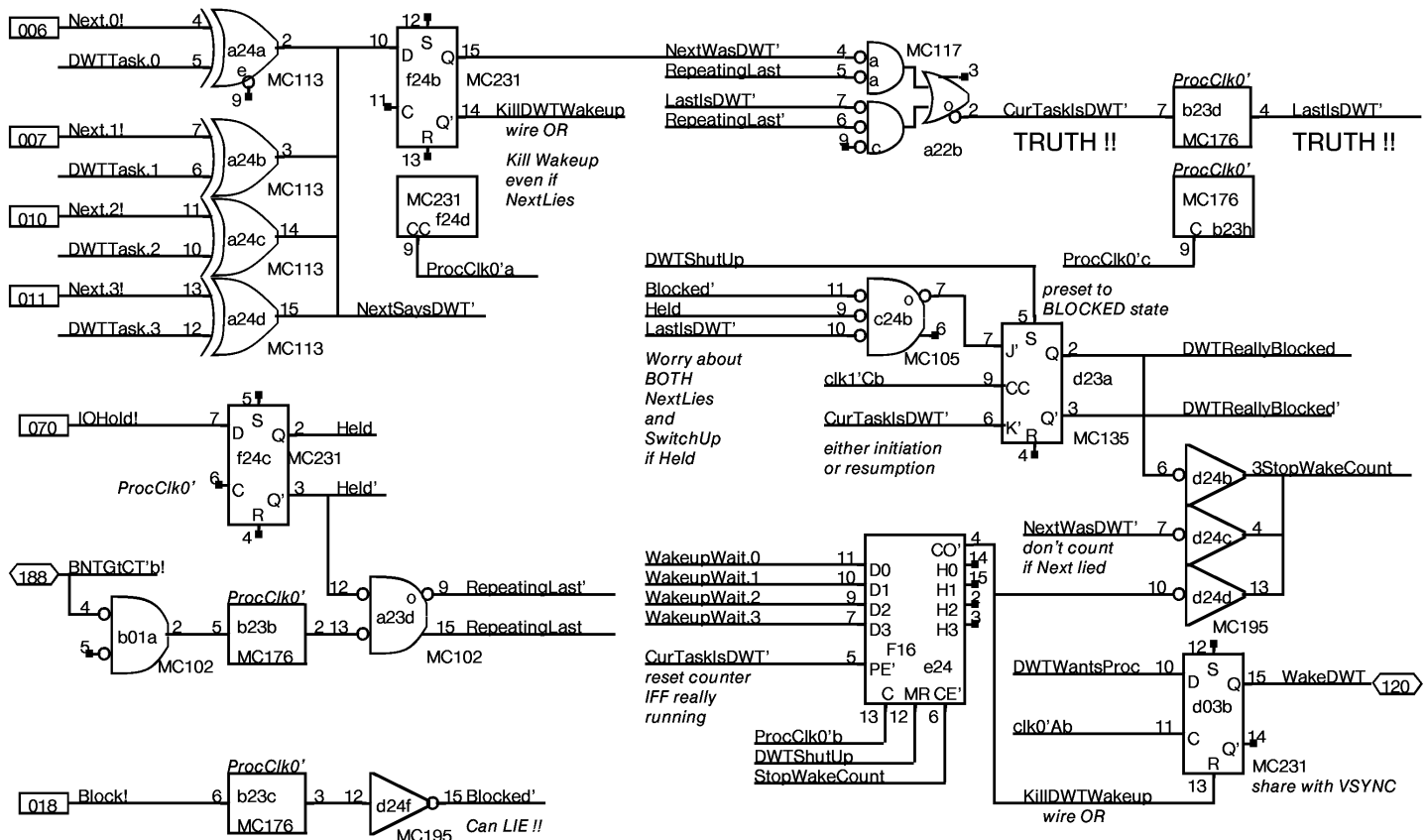


For DWT commands:

RIOB	Means
= 20c	IOFetch signal don't touch Flags
= 1c	Set CWCBFlag and Clear NWCBFlag
= 0c	Clear CWCBFlag

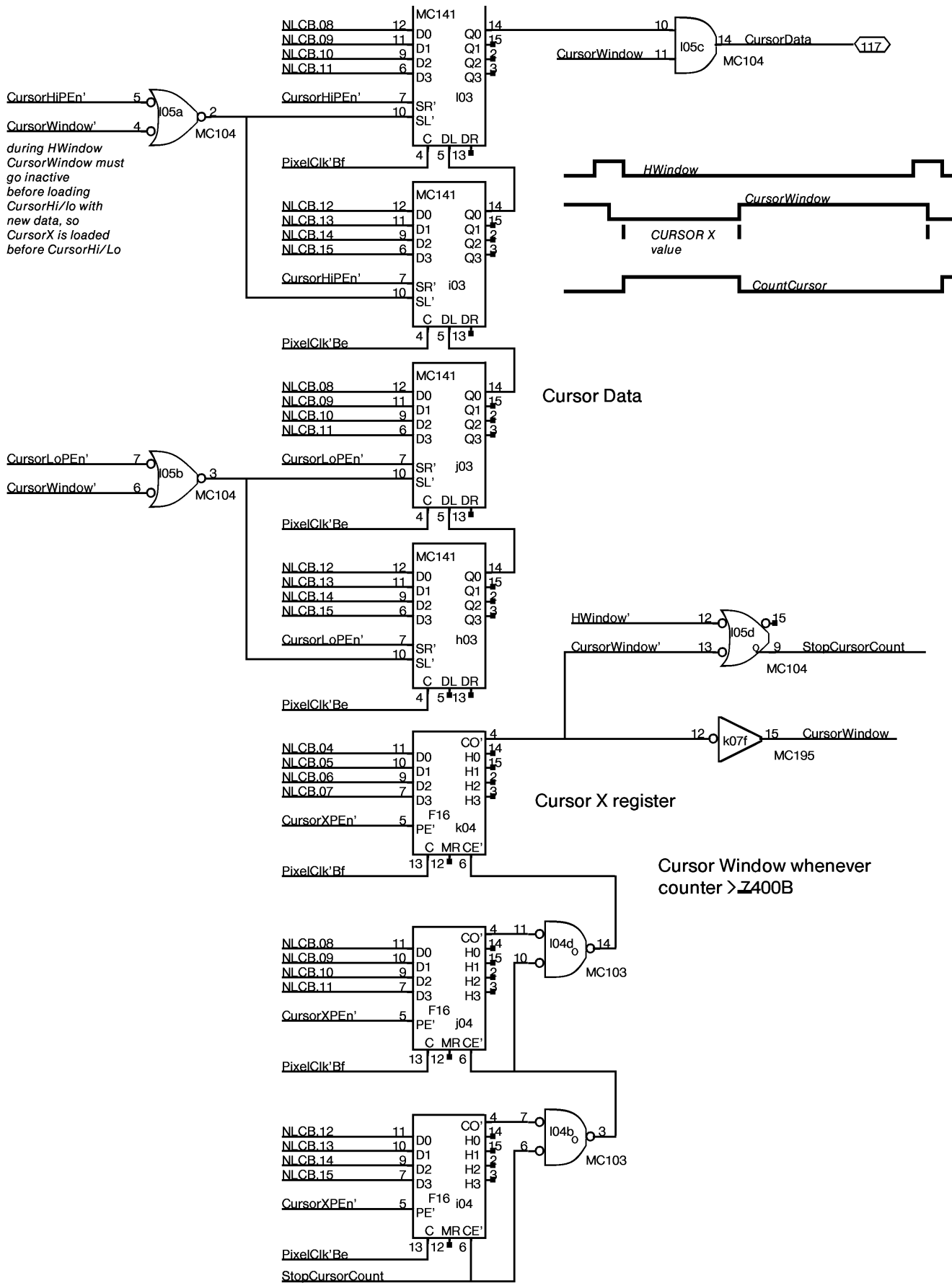
For DHT commands:

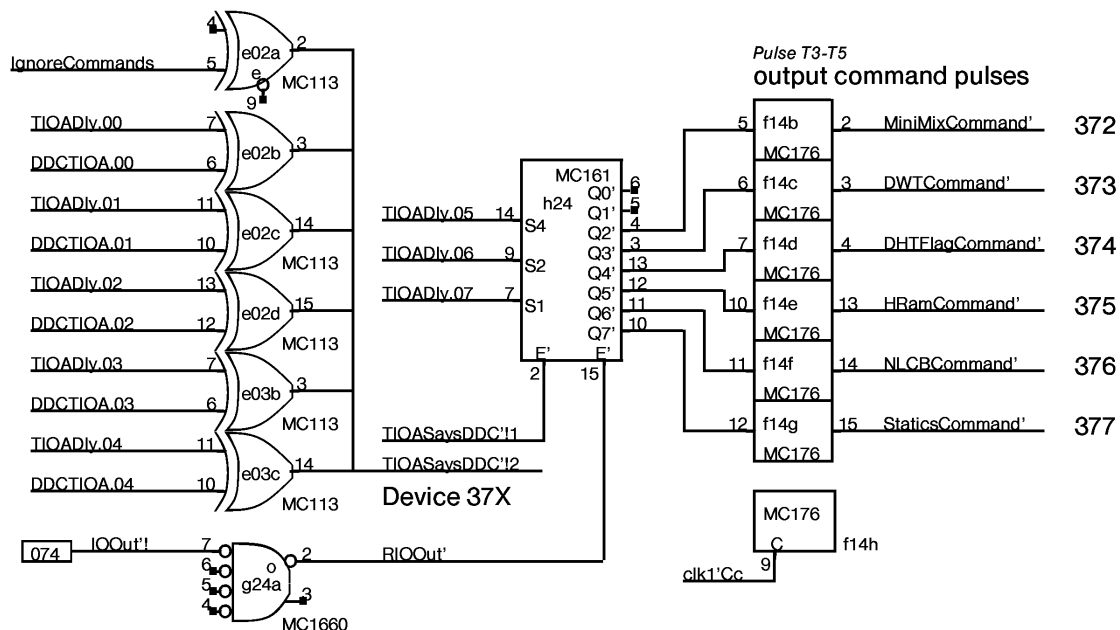
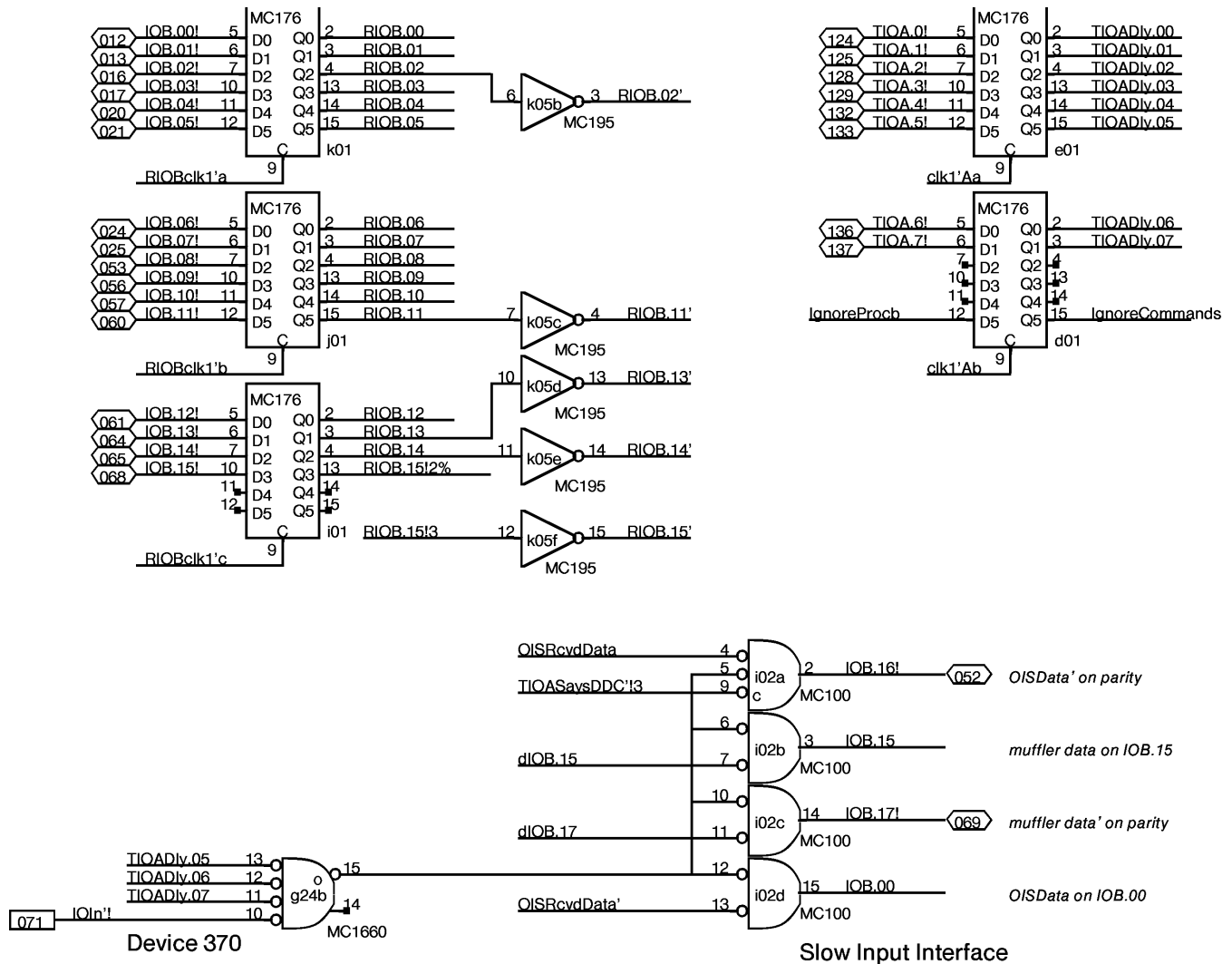
RIOB	Means
= 2c	Set ANextWCBFlag
= 4c	Set BNextWCBFlag

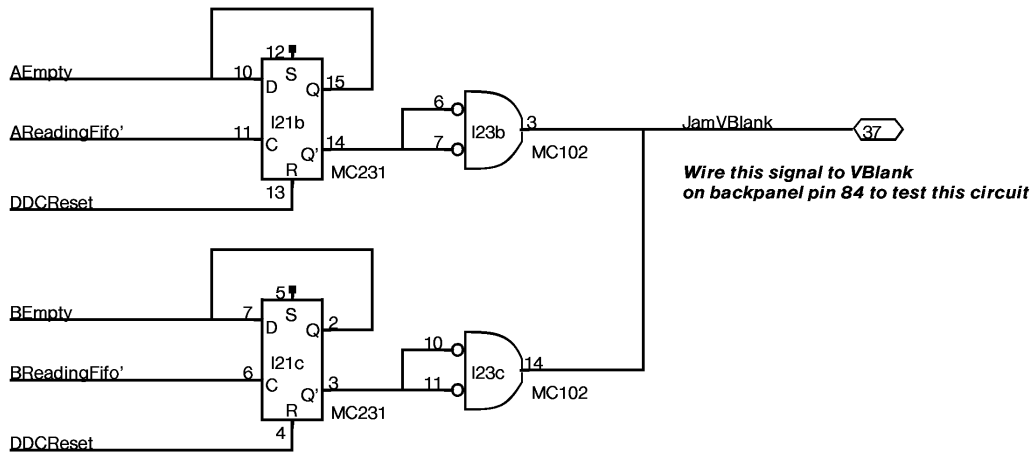
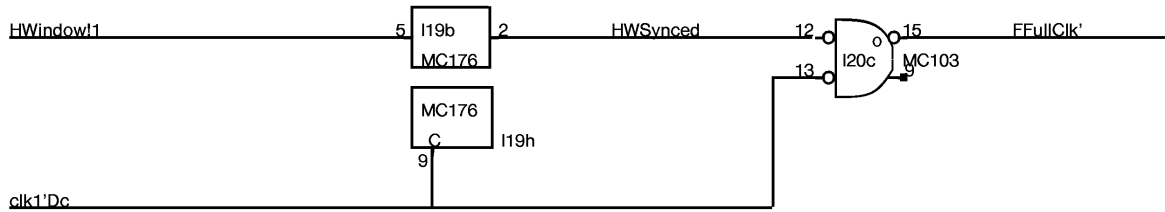
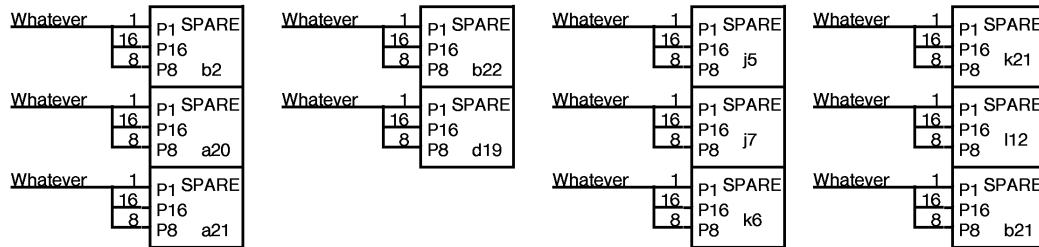
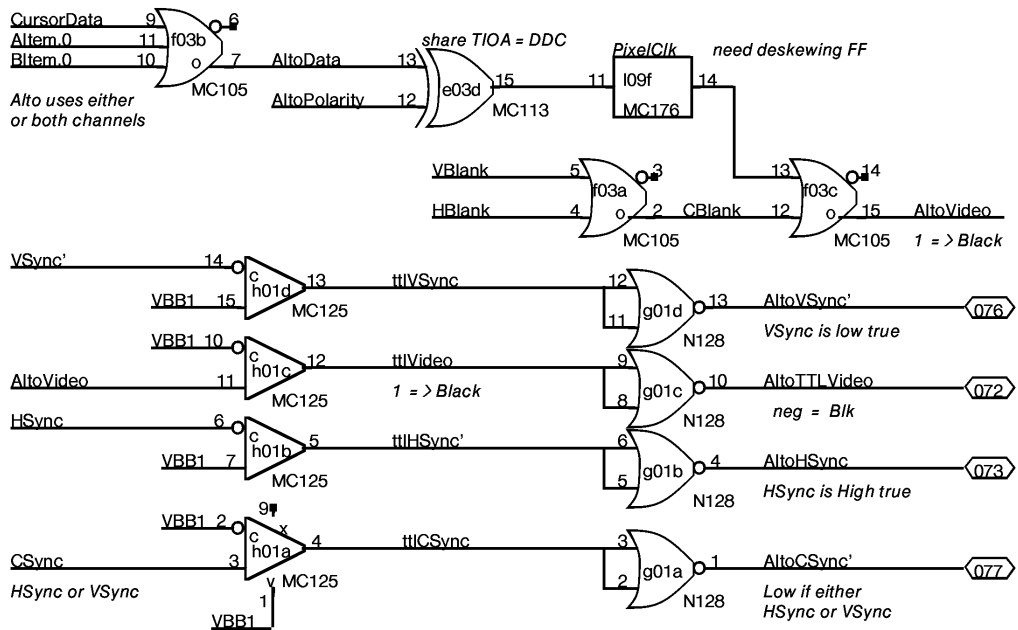


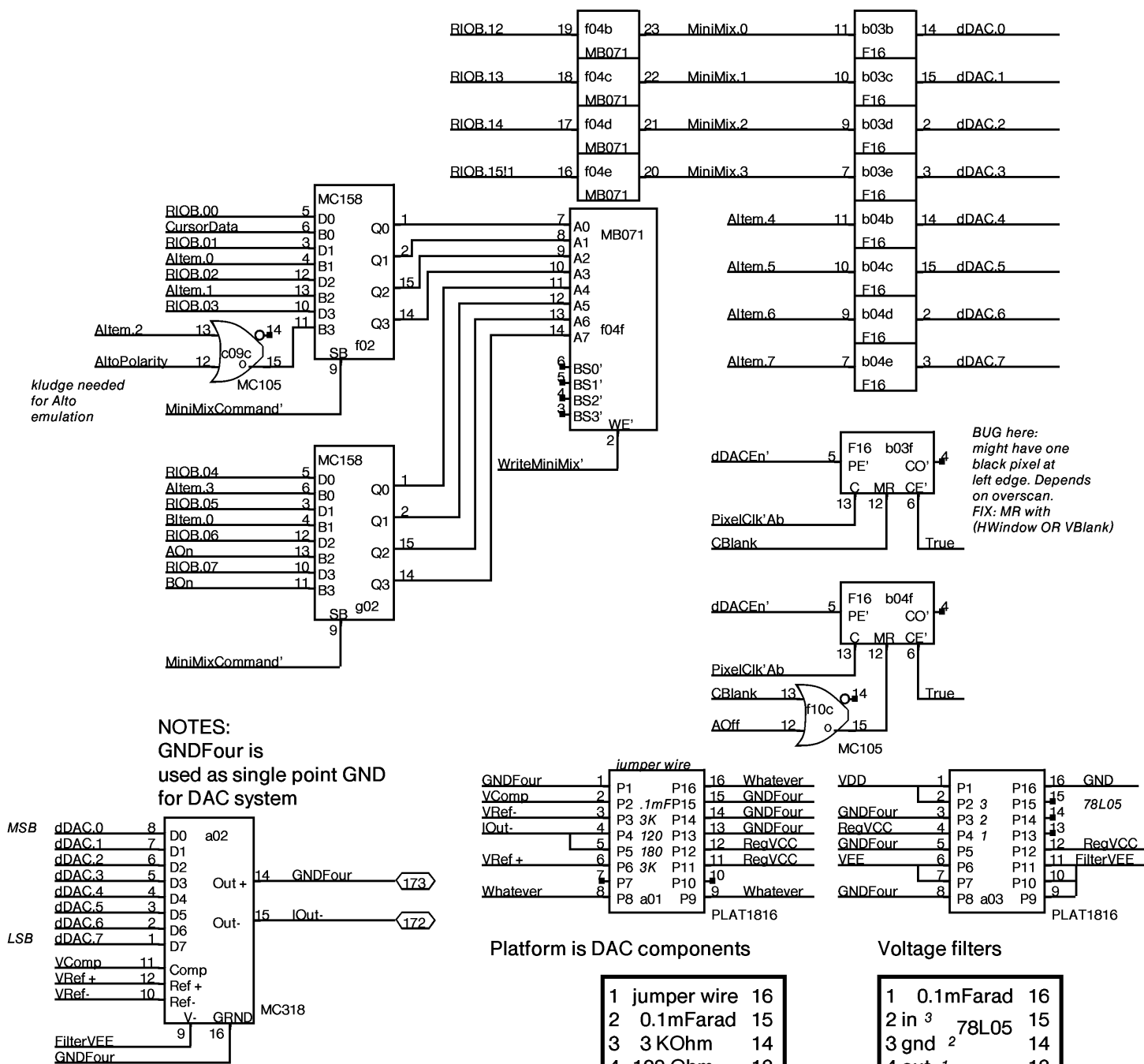
Task Wakeup Requests

all
change
on
clk 1

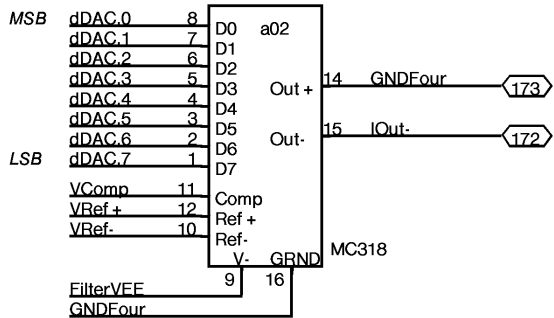








NOTES:
GNDFour is used as single point GND for DAC system



Digital/Analog converter



Platform is DAC components

Voltage filters

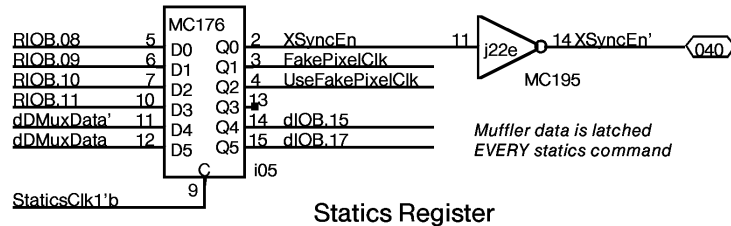
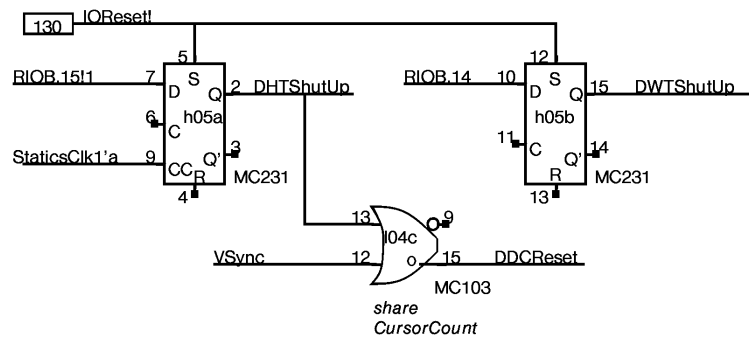
1	jumper wire	16
2	0.1mFarad	15
3	3 KOhm	14
4	120 Ohm	13
5	180 Ohm	12
6	3 KOhm	11
7		10
8		09

location a01

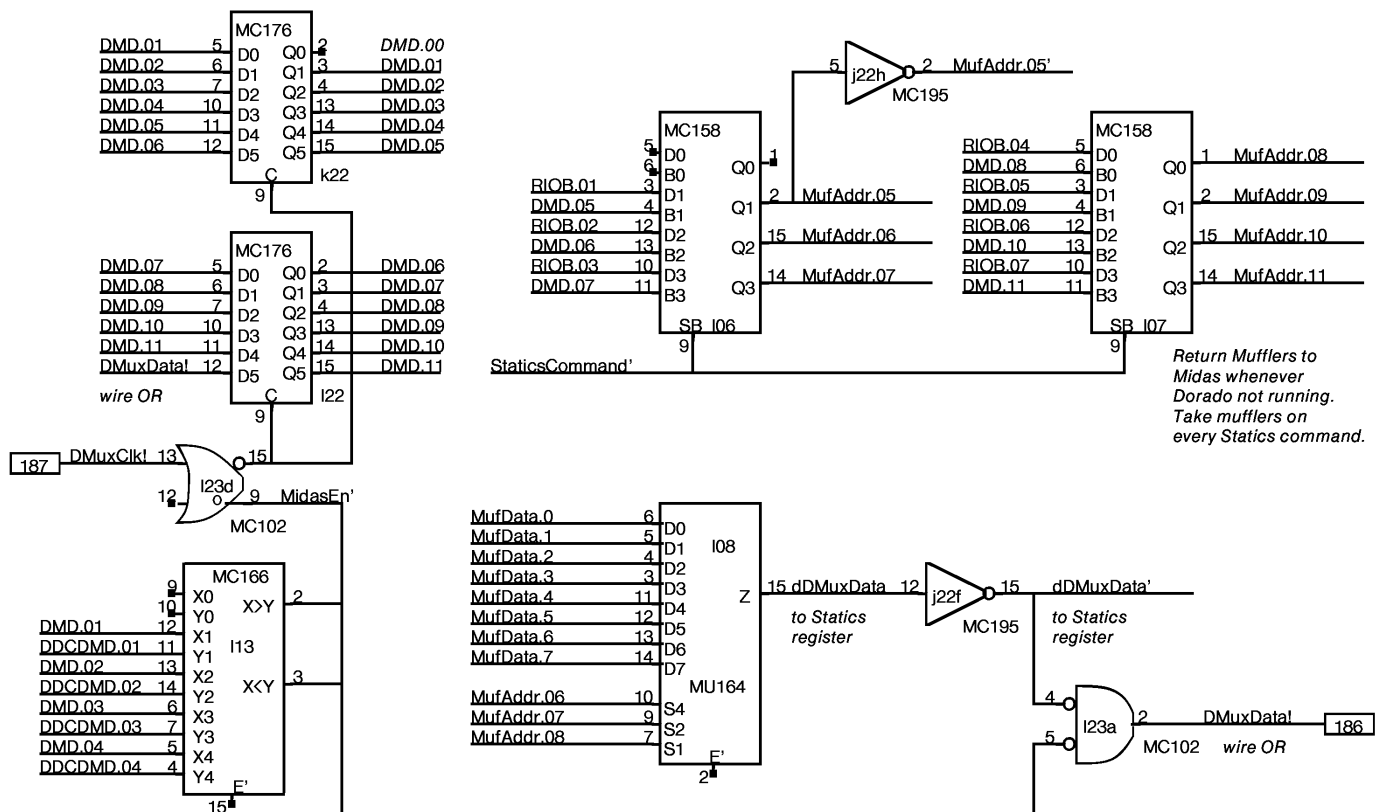
1	0.1mFarad	16
2	in ³	15
3	gnd ²	14
4	out ¹	13
5	0.1mFarad	12
6	12 mHnry	11
7	2 ohm	10
8	+ 22 mF	09

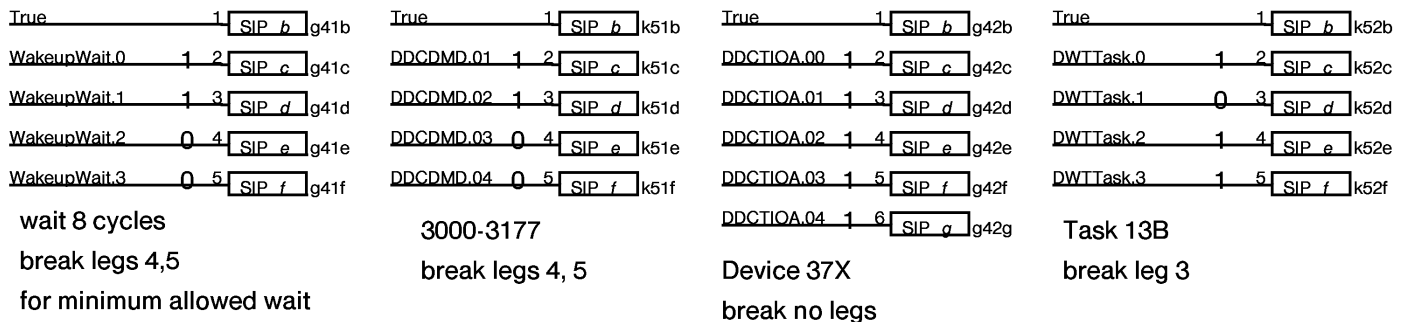
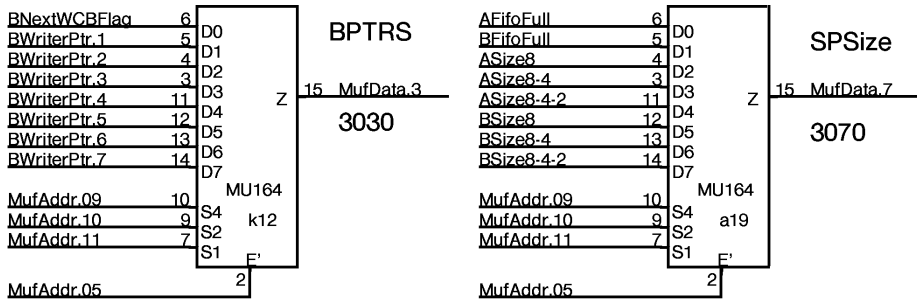
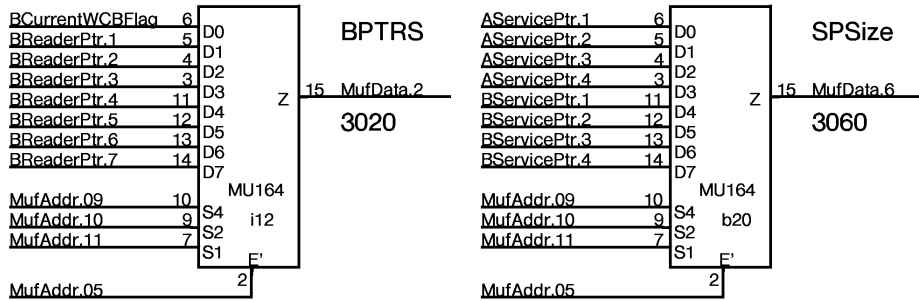
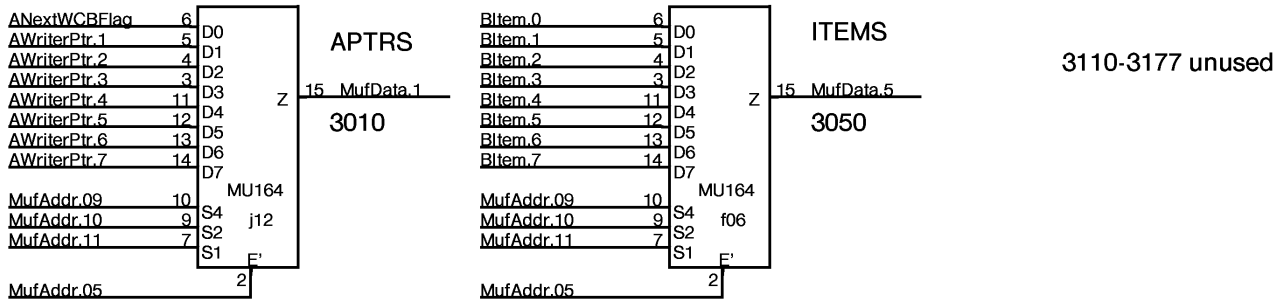
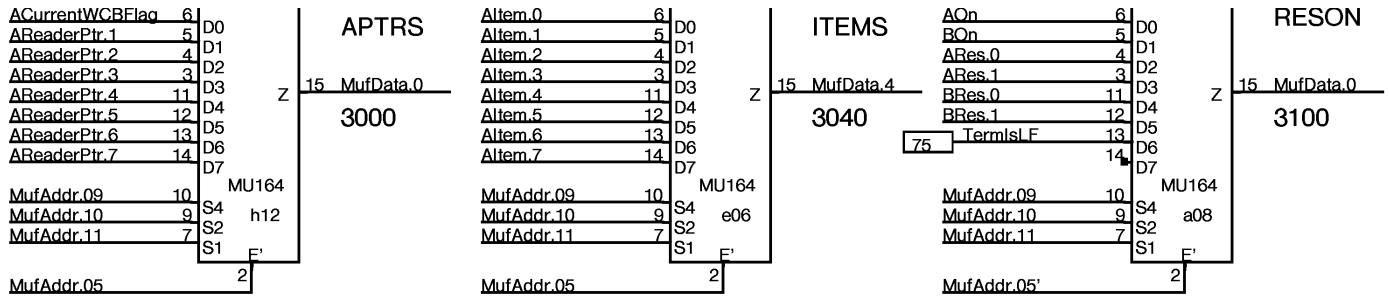
location a03

RevCd: swapped positions of 12mH and 2mF

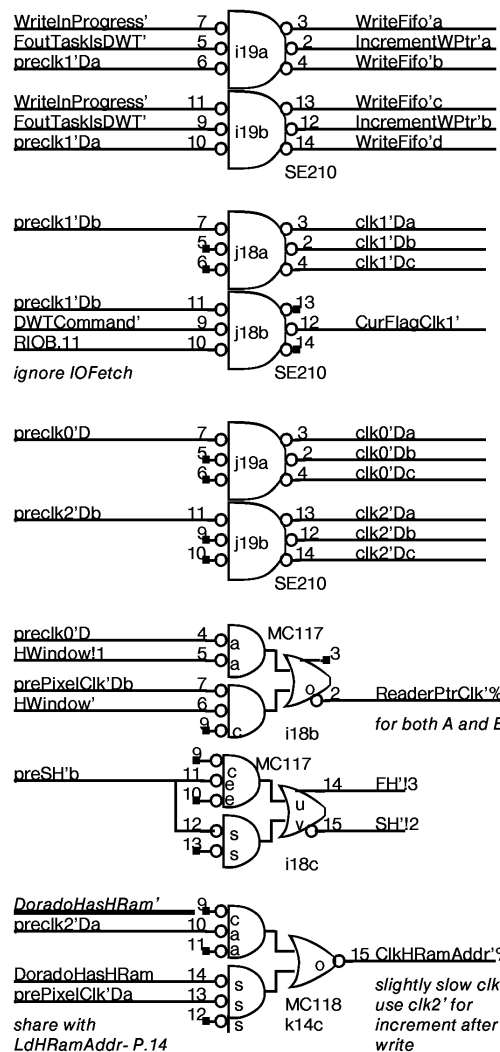
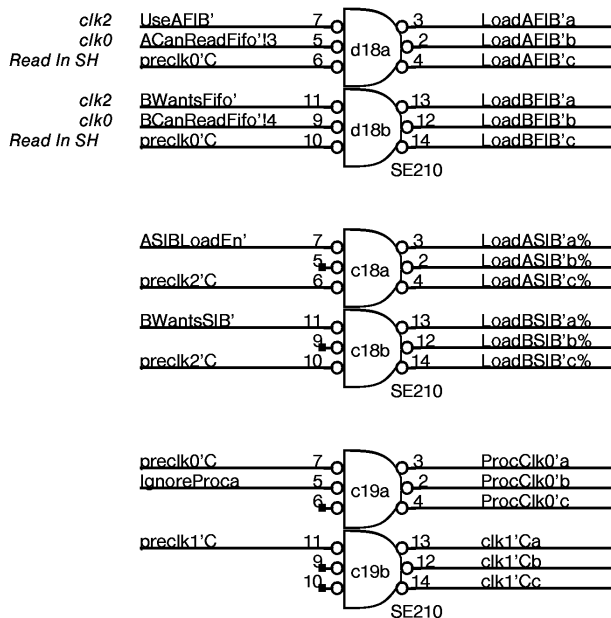
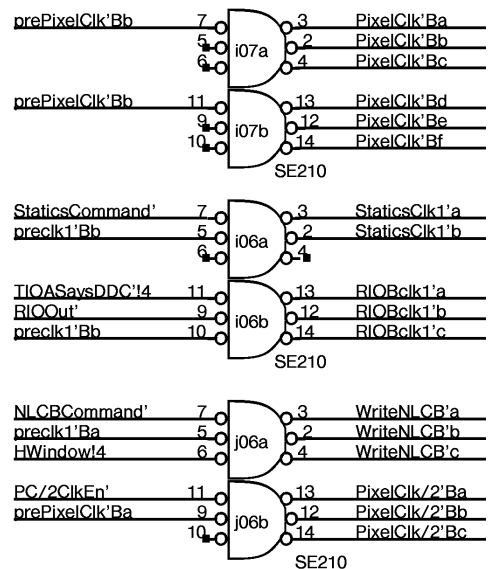
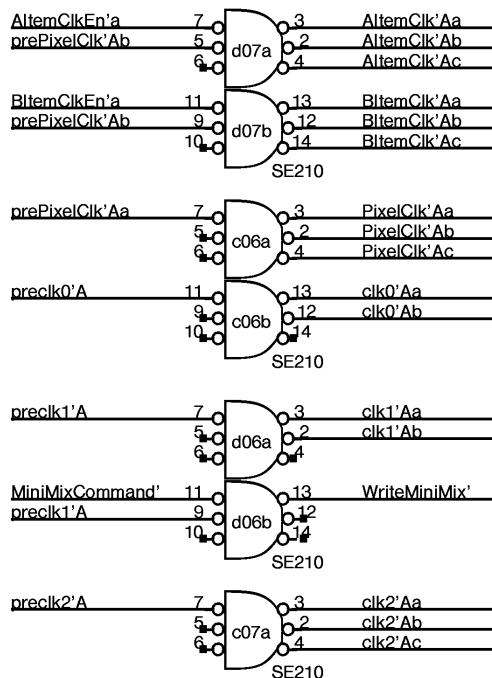


Statics Register

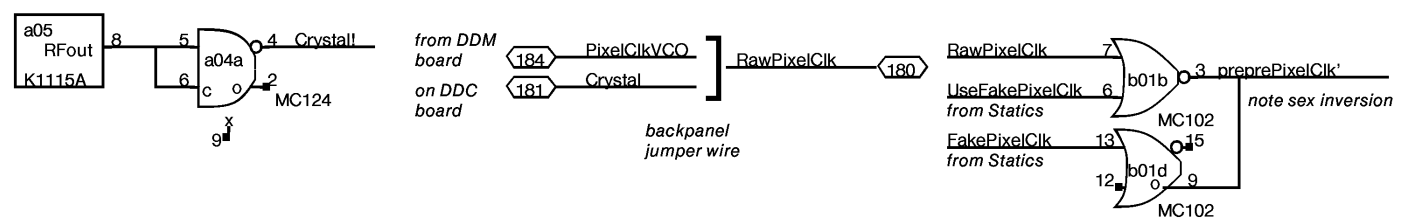
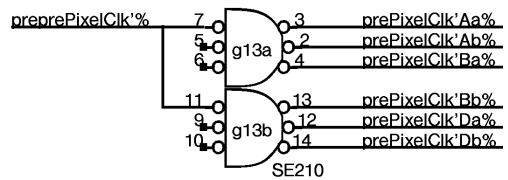
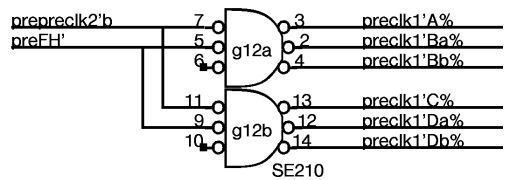
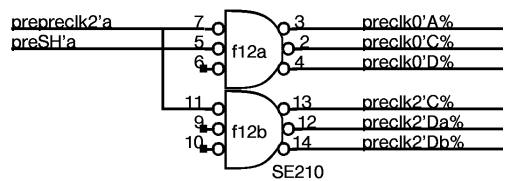
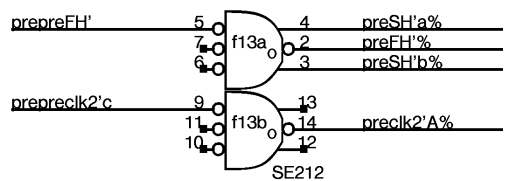
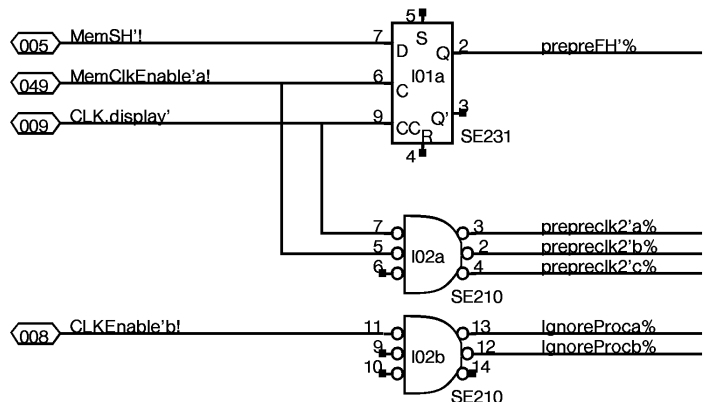




All SIPs are standard 100 ohm terminators with legs broken as shown



CAUTION:
flakey
synchronizer



CONNECTOR		CONNECTOR		TIOA		TWI		CONNECTOR		IOBLo		CONNECTOR		IOBHi		CLK										
A	B>Ct	a	181	b	168	c	153	d	137	e	124	f	109	93	g	80	h	64	i	48	j	33	k	20	l	B
1	41	DAC Plat	PCLK B>Ct 102			135 ASYNC		TIOADEF 176 1		TIOA 176 1		DHTTw RdFifo 135		ALTO 74128		ALTO 125		RIOB 176 rc1		RIOB 176 rc1		RIOB 176 rc1		SE231		1
2		DAC 10318				102 a,b,8,8		OIS 176		TIOA = D 113		MMMux 158		MMMux 158		NLCB 145 W		DIOB 100		NLCB 145 W		NlcbAd F16 P2		SE210		2
3	42	DAC Plat	MMix F16 PC			OIS ICEn 176 PC		DWTTw 231 0		TIOA = D 113		ALTO 105		NLCB 145 W		CURSR 141 P		CURSR 141 P		CURSR 141 P		NlcbAd 197		CURSR 141 P		3
4		PCLK 124 BCL	MMix F16 PC			OIS 105		OIS 101		OIS 176		MMix		ClcbEn 161		VSyC 104		CURX F16 P		CURX F16 P		CURX F16 P		CURX 103		4
5	43	PCLK Xtal	LSR 104			OIS F00		OIS F16		OIS 174		MMix		ClcbEn 161		STAT 231 sc1		STAT 176 sc1				195 RIOB'		CURSR 104		5
6		AdICEN 121	BdICEn 121			PC MMix		0,1		MU AITEM		MU BITEM		ASIZE F16 P2		ARES F16 P2		sc1 rc1		NLCB 1 PC/2				MIDAS 158		6
7	44	ARESC F16 P	BRESC F16 P			2 210 B		ABITEMCLK		AITEM 104		BITEM 104		BSIZE F16 P2		BRES F16 P2		PC PC				PULSE 195		MIDAS 158		7
8		MU ABOFF OIS RES	LSR 118			ASRC F16 I		BSRC F16 I		AITEM 104		BITEM 104		HWND0 135 P2		BWDTH F16 P		BWDTH F16 P		BWDTH F16 P		BWDTH F16 P		MIDAS 164		8
9	45	ASYNC 231 2	ASYNC 231 2			ASYNC 105		BSYNC 231 2		ASYNC 135 I		BSYNC 135 I		MODE F16 P2		ABOFF 135 P		AWDTH F16 P		AWDTH F16 P		AWDTH F16 P		176 PC P		9
10		ASR I F00	BSR I F00			ASR F00		BSR F00		ASR F00		BSYNC 105		VCW F16 P2		BLMarg F16 P		BLMarg F16 P		BLMarg F16 P		BLMarg F16 P		HRmOut F16 P		10
11	46	ASR I F00	BSR I F00			ASR F00		BSR F00		BSR F00		BSR F00		WANT 117		ALMarg F16 P		ALMarg F16 P		ALMarg F16 P		ALMarg F16 P		AWDTH 103		11
12		ASR I F00	BSR I F00			ASR F00		BSR F00		ASR F00		pc0' pc2' SE210		pc1' pc1' SE210		MU ARP AF		MU BRP BF		MU AWP AF		MU BWP BF				12
13	47	ASIB 176	BSIB 176			AFIB F00		BFIB F00		BFIB F00		pc2 FHS SE212		ppc' ppc' SE210		HRAddr F16 H		HRAddr F16 H		HRAddr F16 H		HRAddr F16 H		MIDAS 166		13
14		ASIB 176	BSIB 176			AFIB F00		AFIB F00		BFIB F00		COMM' 176 1		flags 117		HRam 415 HW		HRam 415 HW		HRam 415 HW		HRamC 118		HRamC 118		14
15	48	ASIB 176	BSIB 176			AFIB F00		BFIB F00		IP		NFlags 231 1		CFlags 231 cf		DHasHR 231 1,p2		AFULL 149		BFULL 176 1		AFULL 176 1		BFULL 149		15
16		ASIB 176	BSIB 176			AFIB F00		BFIB F00		IP		IP						ARP rpc F16		BRP rpc F16		AWP wpc F16		BWP wpc F16		16
17	49	ASIB 176	BSIB 176			AFIB F00		BFIB F00		IP		IP		FIFO		FIFO		ARP rpc F16		BRP rpc F16		AWP wpc F16		BWP wpc F16		17
18		ASIB 176	BSIB 176			ABSIB 2 ABSIB 2		ABFIB 0		IP		IP						FHS rpc 0 117		1, cflg 1		FAM 174		PTRS 103		18
19	50	MU BA-SZE	ABSize 103			PROCD' 1'				IP		IP		FIFO		FIFO		fifo 1 wpc 1		2 0		FAM 174		Sync 176		19
20			MU SP			AFIB F00		BFIB F00		IP		IP						FA 2 231		FAM FH 174 SH		FAM 174		IncSP 103 c		20
21	51					AFIB F00		BFIB F00		IP		IP		FIFO		FIFO		FA 2 231		FA 2 231						21
22		CTDWT 117 c				OIS 231		IP		IP		IP						FA 2 231		d 195		MIDAS 176		MIDAS 176		22
23	52	102 c	clk0 176 FG			GETS 231 1		BLKD B 135 1		ASP 1 F16		BSP 1 F16		FIFO		FIFO		FOUT 1 176		FOUT 1 176		FOUT 1 176		MIDAS 102 BC		23
24		NEXT 113	SUBT 106 A			105		195		WSpace F16 0		HELD 231 0		IOInOut 1660		TIOA = D 161		FOUT 0 176		FOUT 0 176		FOUT 0 176 f		FTsk 113		24
C		a	11	b	26	c	39	d	55	e	70	f	86	99	g	114	h	129	i	143	j	159	k	174	l	D
E		NEXT	SubT	BLK				FIN		Hold		IOIn/out		CONNECTOR		IOB FNXT		FOUT				FTsk		DMux		E
XEROX PARC		Project Dorado		Reference DDC Board Layout				File DispY26.sil		Designer K. Pier		Rev CI	Date 3/25/82	Page 26												

DDC Slow IO System

DEVICE	TIOA	I/O	TASK	FORMAT and COMMENTS
STATICS	377	O	DHT, EMU	See Below
NLCB	376	O	DHT, EMU	NLCBAddr.0-3,,Data.4-15
HRAM	375	O	DHT, EMU	Keep,Write,LoadAddr,0,Data.4-15
DHTFLAG	374	O	DHT	,ANextWCBFlagSet,BNextWCBFlagSet,0 bits 13,14,15
DWT	373	O	DWT	IOFetch signal,....,Set/Clr CWCBCFlagANDClrNWCBCFlag bit 11, , bit 15
MiniMixer	372	O	DHT, EMU	address,,data always writes MiniMixer RAM
STATUS	370	I	DHT, EMU	Selected muffler input returned in bit 15
PIXELCLK	367	O	DHT, EMU	Pixel clock rate located on mixer board
MIXER	366	O	DHT, EMU	Keep,Write,LoadAddr,0,Data.4-15 located on mixer board

00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
X	Muff Addr .05	Muff Addr .06	Muff Addr .07	Muff Addr .08	Muff Addr .09	Muff Addr .10	Muff Addr .11	XSync Enable	Fake Pixel Clock	Use Fake Pixel Clock	Unused	Unused	Unused	DWT Shut Up	DHT Shut Up-Reset DDC	Statics
NLCB Addr .00	NLCB Addr .01	NLCB Addr .02	NLCB Addr .03	NLCB Data 12 bits												NLCB
Keep HRam'	Write HRam'	Load Hram Addr				Addr.0	Addr.1	Addr.2	Addr.3	Addr.4	Addr.5	Addr.6	Addr.7	Addr.8	Addr.9	HRAM
														HSync	HBlank	
													Set BNext WCB Flag	Set ANext WCB Flag	Must Be 0	DHTFLAG
											IOFetch signal	Must Be 0	Must Be 0	Must Be 0	Set/Clr Cur WCB Flag	DWT
Address.0-7								Data.0-7								MiniMixer
								ClkRate.0-7								PIXELCLK
Keep Mixer'	Write Mixer'	Load Mixer Addr	X		Addr.0	Addr.1	Addr.2	Addr.3	Addr.4	Addr.5	Addr.6	Addr.7	Addr.8	Addr.9	Hi/Lo select	MIXER
				Mixer Data 12 bits												

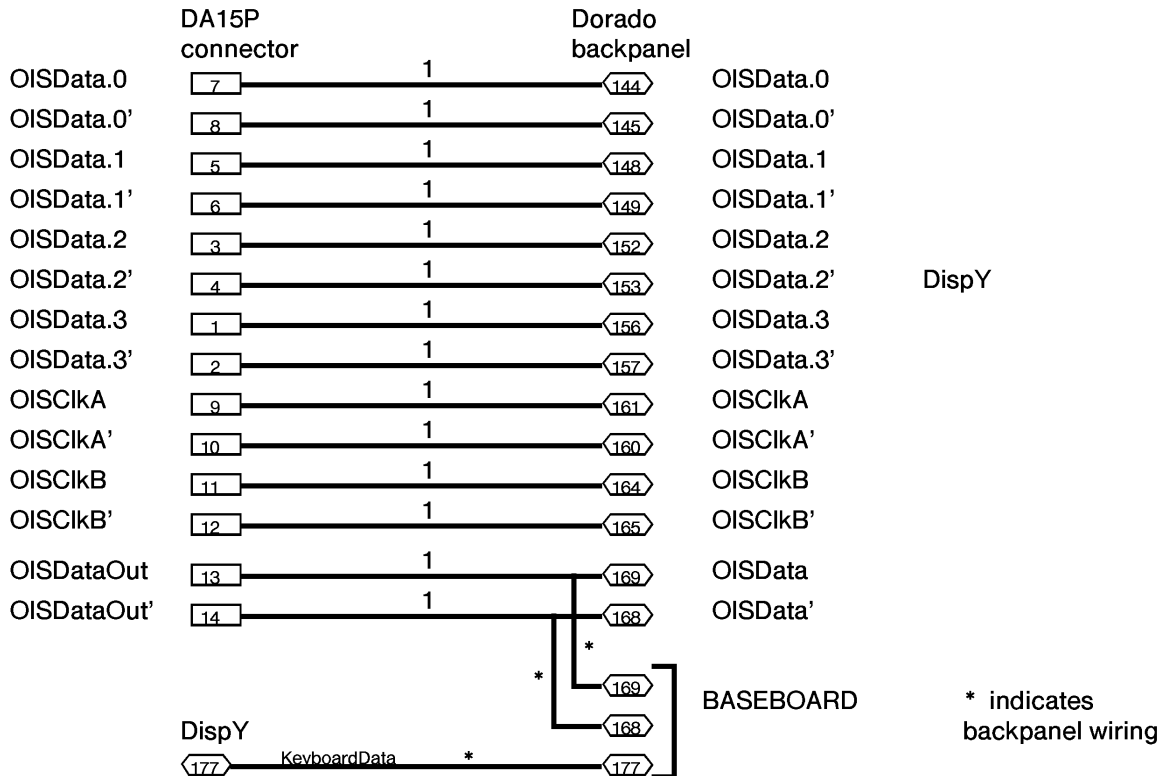
XEROX PARC	Project	Reference	File	Designer	Rev	Date	Page
	Dorado	DDC Slow IO System Formats	DispY27.sil	K. Pier	CI	7/19/79	27

Next Line Control Block Format

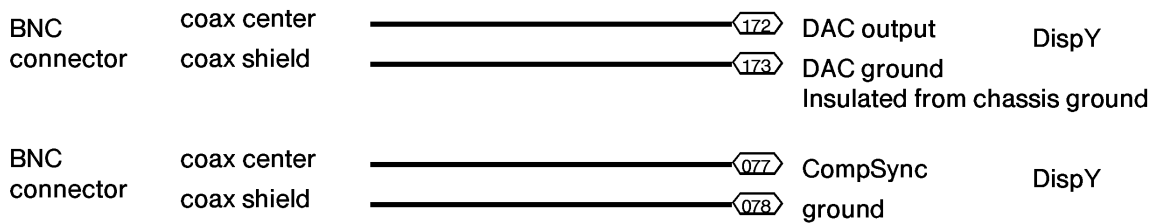
Address	Name	Format
0	VCW vertical control word	0...0, VBlank, VSync, EvenField
1	AMargin	LMarg[00..11]
2	AWidth	Width[00..11]
3	AFifoAddr	FifoAddr[0..7] *must be even
4	AScan	0...0,Polarity,Resolution[0..1],Size8,Size4,Size2,Size1
5	ModeControl	0..0,24Bit,ABypass,BBypass,A8B2
6	reserved	
7	reserved	
10	unused	
11	BMargin	LMarg[00..11]
12	BWidth	Width[00..11]
13	BFifoAddr	FifoAddr[0..7] *must be even
14	BScan	0...0,Polarity,Resolution[0..1],Size8,Size4,Size2,Size1
15	CursorX	CursorXCount[0..11]
16	CursorLo	CursorLoByte[4..11]
17	CursorHi	CursorHiByte[4..11]

DDC	SIZE	DDM
Altem	8	Altem 85-88-89-92-93-96-97-100
Bltem	8	Bltem 101-104-105-108-109-112-113-116
CursorData	1	CursorData
AltemClkEn	1	AltemClkEn
BltemClkEn	1	BltemClkEn
AOff	1	AOff
BOff	1	BOff
HSync	1	HSync
HBlank	1	HBlank
HalfLine	1	HalfLine
VSync	1	VSync
VBlank	1	VBlank
PixelClk	1	PixelClk
Crystal	1	Crystal
Oscillator	1	Oscillator
Modes	3	Modes 28-32-33
XHSync	1	XHSync
XVSync	1	XVSync
XSyncEn	1	XSyncEn
OISData.0	1	7 wire interface cable
OISData.0'	1	
OISData.1	1	
OISData.1'	1	
OISData.2	1	
OISData.2'	1	
OISData.3	1	
OISData.3'	1	
OISClkA	1	
OISClkA'	1	
OISClkB	1	
OISClkB'	1	
OISDataOut	1	
OISDataOut'	1	
AltoVideo	1	Alto monitor cable
AltoHSync	1	
AltoVSync	1	
AltoCSync	1	Grey level driver cables 2 coax
DAC	1	
DAC GND	1	
spare	1	from Baseboard
KeyboardData	1	

Seven Wire Interface Cable



Grey level cabling



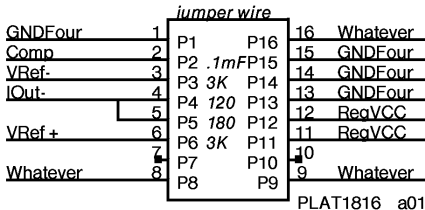
Note: CompSync currently appears in the Alto cable. The Alto display doesn't make use of CompSync. We could simply replace the CompSync pair in the Alto cable with the coax. We then have the capability to attach a local Alto type display and a remote display.

Six BNC connectors mounted on the "square" side. Cabling not yet defined.

1. LEAVE LOCATION a02 BLANK. Do not plug in MC10318.
2. Plat1816 in locations a01 and a03 are discrete components shown below.
3. SIPs are 100 ohm terminator package with legs broken:

location	break legs	See page 23.
g41	4,5	
g42	none	
k51	4,5	
k52	3	

4. SIPs at locations d42 and e42 are 220 ohm value instead of 100 ohm.
5. Crystal oscillator K1115A, location a05, value 50 mhz

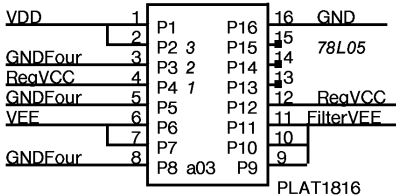


Platform is DAC components

1	jumper wire	16
2	0.1mFarad	15
3	3 KOhm	14
4	120 Ohm	13
5	180 Ohm	12
6	3 KOhm	11
7		10
8		09

location a01

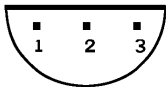
See Page 20.



Voltage filters

1	0.1mFarad	16
2	in ³	15
3	gnd ²	14
4	out ¹	13
5	0.1mFarad	12
6	12 mHnry	11
7	2 ohm	10
8	+ 22 mF	09

location a03



1. Output
2. Ground
3. Input

Bottom view of pinout for 78L05